Software Requirements Specification

for

VendorSetu

Street Vendor Formalization & Micro-Credit Platform

Version 1.0 approved

Prepared by: Nancy Yadav, <Member 2>, <Member 3>, <Member 4>

B.Tech CSE, Batch 2024 - CSE3005 Software Engineering

BML Munjal University

Submitted to: <Faculty name>

28 October 2026

Table of Contents

Table of Contents 2
Revision History 5
1. Introduction 5
1.1 Purpose 5
1.2 Document Conventions 5
1.3 Intended Audience and Reading Suggestions 5
1.4 Product Scope 6
1.5 References 6
2. Overall Description 7
2.1 Product Perspective 7
2.2 Product Functions 7
2.3 User Classes and Characteristics 7
2.4 Operating Environment 8
2.5 Design and Implementation Constraints 8
2.6 User Documentation 8
2.7 Assumptions and Dependencies 8
3. External Interface Requirements 9
3.1 User Interfaces 9
3.2 Hardware Interfaces 9
3.3 Software Interfaces 9
3.4 Communications Interfaces 10
4. System Features 10
4.1 User Registration & Authentication [Implemented] 10
4.2 Vendor Profile & Daily Sales Logging [Implemented] 11
4.3 Alternative Credit (Trust) Scoring Engine [Implemented] 11
4.4 Loan Application & Progressive Lending [Implemented] 12
4.5 Loan Review & Disbursal (Bank / NBFC) [Implemented] 12
4.6 EMI Repayment Tracking [Implemented] 13
4.7 Vending Zone Licensing 13
4.8 Dispute / Eviction Case Logging 14
4.9 Customer: Locate, Pay & Rate Vendors 14
4.10 Peer-Guarantor Group Lending 15
5. Other Nonfunctional Requirements 15

5.1 Performance Requirements 15
5.2 Safety Requirements 15
5.3 Security Requirements 16
5.4 Software Quality Attributes 16
5.5 Business Rules 16
6. Other Requirements 16
6.1 Database Requirements 16
6.2 Legal and Regulatory Requirements 17
6.3 Internationalisation Requirements 17
6.4 Reuse Objectives 17
Appendix A: Glossary 17
Appendix B: Analysis Models 18
B.1 Use Case Diagram 18
B.2 Class Diagram 20
B.3 Sequence Diagrams 21
B.4 State Transition Diagram 24
B.5 Data Flow Diagrams 25
Appendix C: To Be Determined List 28
Appendix D: Implementation (Release 1.0 Prototype) 28
D.1 Development Approach and Technology 28
D.2 Coded Modules 28
D.3 Project Structure 28
D.4 Key Algorithms 29
D.5 How to Run 30
D.6 Coding Standards Followed 30
Appendix E: Test Cases and Results 31
E.1 Test Strategy 31
E.2 User Registration & Login 32
E.3 Vendor Profile & Daily Sales Logging 34
E.4 Trust Score Calculation 36
E.5 Loan Application & Bank Review 38
E.6 EMI Schedule & Repayment 40
E.7 Test Summary 42
E.8 Defects Found by Testing 42

Revision History

Name	Date	Reason For Changes	Version
Nancy Yadav & team	28-10-2026	Complete SRS for final evaluation: requirements, analysis models (UML & DFD), prototype implementation of 4 modules and executed test cases	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of **VendorSetu, release 1.0** - a web platform that gives India's street vendors a digital business identity, records their daily sales, converts that history into an alternative credit **Trust Score**, and uses it to offer progressive micro-loans modelled on the Government of India's PM SVANidhi scheme. The platform also digitises vending-zone licensing for municipal bodies and lets customers find and pay vendors.

The SRS covers the complete system and all five user classes. Features marked **[Implemented]** are built and tested in the release-1.0 prototype submitted with this document (Appendix D and E); the remaining features are fully specified here and scheduled for the next increment.

1.2 Document Conventions

- **Requirement IDs:** functional requirements are tagged FR-<feature>-<n> (e.g. FR-LN-3), non-functional requirements NFR-<category>-<n>, and business rules BR-<n>. IDs are referenced in the source code comments.
- **Priorities:** every feature carries a MoSCoW priority (Must / Should / Could / Won't) and a High / Medium / Low rating. Detailed requirements inherit the priority of their feature unless stated otherwise.
- **"shall"** marks a mandatory requirement; **"should"** marks a desirable one.
- **[Implemented]** marks features coded in the release-1.0 prototype.
- Amounts are in Indian Rupees (₹). Dates are written DD-MM-YYYY.
- Items that are not yet decided are marked **TBD-n** and tracked in Appendix C.

1.3 Intended Audience and Reading Suggestions

Reader	Interest	Suggested sections
Course faculty / evaluators	Completeness and correctness of the SRS, models and testing	1, 2, 4, Appendix B, D, E
Developers	What to build and the business rules	2, 3, 4, 5.5, Appendix B, D
Testers	Verifiable requirements and expected behaviour	4, 5, Appendix E
Bank / NBFC partners	Loan rules, trust score, repayment and default handling	1.4, 4.3-4.6, 5.3, 6.2
Municipal (ULB) officials	Licensing and dispute features	2.3, 4.7, 4.8
Project manager	Scope, priorities, assumptions and open issues	1.4, 2.7, 4 (MoSCoW), Appendix C

Recommended order: read Section 1 and 2 for the overview, then Section 4 for the features, Appendix B for the diagrams, Section 5 for quality requirements, and Appendix D-E for the prototype and its test results.

1.4 Product Scope

Street vendors are a large part of India's informal economy, yet most have no formal credit history. Banks cannot assess them with a CIBIL score, so vendors depend on informal moneylenders. PM SVANidhi showed that small, progressive working-capital loans work - the restructured scheme (approved by the Union Cabinet in August 2025, extended to March 2030) targets 1.15 crore beneficiaries with an outlay of ₹7,332 crore - but application, verification and repayment tracking are still largely manual.

VendorSetu's objectives are to:

- give every vendor a digital profile and an easy daily sales log (cash or UPI);
- compute a transparent **Trust Score (0-100)** from sales consistency, repayment history, peer vouching and vending tenure, so lenders can assess vendors without a credit bureau history;
- offer **progressive loans** of ₹15,000 → ₹25,000 → ₹50,000, where repaying one tier unlocks the next;
- let bank/NBFC officers review, approve, disburse and monitor loans with a generated EMI schedule;
- digitise vending licences and dispute/eviction cases for municipal officers;
- help customers locate nearby vendors, pay digitally and rate hygiene and service.

Benefits: faster and fairer credit decisions, lower default risk through progressive limits, a verifiable business record that helps vendors formalise, and better data for urban local bodies.

1.5 References

#	Document	Author / Source	Version / Date
R1	IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830)	IEEE	1998
R2	Software Requirements Specification template	Karl E. Wiegers	1999
R3	PM SVANidhi scheme portal and guidelines (pmsvanidhi.mohua.gov.in)	Ministry of Housing & Urban Affairs, GoI	Accessed Oct 2026
R4	Cabinet approves restructuring and extension of PM SVANidhi Scheme till 2030	Union Cabinet (news reports)	27-08-2025
R5	Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act	Government of India	2014
R6	Guidelines on Digital Lending	Reserve Bank of India	02-09-2022
R7	Digital Personal Data Protection Act	Government of India	2023
R8	Software Engineering, 9th edition	Ian Sommerville, Pearson	2011
R9	Software Engineering: A Practitioner's Approach, 8th edition	Roger S. Pressman, McGraw Hill	2014
R10	CSE3005 Software Engineering - Course Handout	BML Munjal University	2026-27
R11	Test Case Template (Doc T_01)	Dr Ghazy Assassa	-

2. Overall Description

2.1 Product Perspective

VendorSetu is a **new, self-contained product**. It does not replace the PM SVANidhi portal; it complements it by producing the evidence a lender needs (a verified sales record and a Trust Score) and by tracking the loan through its whole life cycle. Today a vendor needs a vending certificate or a Letter of Recommendation from the urban local body, and then bank paperwork; VendorSetu brings the vendor, the municipal officer and the lender onto one platform.

Major components and external interfaces (see the Context Diagram, Figure 6):

- **Web client** - responsive pages used from a smartphone or PC browser by all user classes.
- **Application server** - Python/Flask application with four business modules: Authentication, Vendor Profile & Sales, Trust Score Engine, Loan Management.
- **Database** - relational store for users, profiles, sales, vouches, loans and EMI installments.
- **External systems** - UPI/payment gateway (EMI and customer payments), SMS/OTP gateway (TBD-4), and the lender's own core banking system, which performs the actual fund transfer.

2.2 Product Functions

- **All users:** register, log in with mobile number and password, role-based dashboards.
- **Street vendor:** maintain a business profile; log daily sales; view a 30-day summary and Trust Score with its breakdown; apply for a loan within the eligible tier; view the EMI schedule and pay EMIs; apply for/renew a vending licence; view the vending-zone map.
- **Trust Score Engine:** compute a 0-100 score and risk band; decide loan eligibility (score ≥ 40).
- **Bank / NBFC officer:** review applications with Trust Score, approve or reject with remarks, disburse and generate the EMI schedule, track repayments and mark defaults.
- **Municipal officer:** approve/renew vending licences, log dispute or eviction cases, view the vendor registry.
- **Customer:** locate vendors by area, pay digitally, rate hygiene and service.
- **Peer guarantor:** vouch for another vendor (raises their Trust Score) and join group-lending circles.

2.3 User Classes and Characteristics

User class	Characteristics	Usage / privileges	Importance
Street Vendor	Small business owner; basic smartphone skills; may have limited literacy and prefer Hindi or a regional language	Daily (sales log); monthly (EMI). Own data only	Primary - most important
Bank / NBFC Officer	Trained loan officer of a regulated lender; comfortable with web forms	Daily. Reviews and disburses loans; sees vendors' loan data	Primary
Municipal Officer	Urban local body / Town Vending Committee staff	Weekly. Licences and dispute cases within own ward	Secondary
Customer	General public with a smartphone	Occasional. Search, pay, rate	Secondary
Peer Guarantor	A registered vendor who vouches for another vendor	Occasional	Secondary
System Administrator	Technical staff	Maintenance, backups, user unlock	Support

2.4 Operating Environment

- **Server:** any OS that runs Python 3.11+ (developed on Windows 11; deployable on Linux). Flask 3.1, Werkzeug, Jinja2.
- **Database:** SQLite 3 in the prototype; PostgreSQL 15 recommended for production.
- **Client:** current Chrome, Edge, Firefox or Safari; Android 8+ smartphones on 3G/4G networks.
- **Network:** HTTPS over the public internet.

2.5 Design and Implementation Constraints

- **Regulatory:** loans must be sanctioned and disbursed by an RBI-regulated entity directly to the borrower's bank account (RBI Digital Lending Guidelines, R6); VendorSetu records decisions but does not hold funds.
- **Privacy:** personal data must be processed with consent and minimised (DPDP Act 2023, R7); full Aadhaar numbers shall not be stored.
- **Devices:** pages must work on low-end Android phones and slow networks (see NFR-PERF-3).
- **Architecture:** business logic is kept in UI-independent modules so it can be unit-tested and reused by a future mobile app or API.
- **Technology (prototype):** Python 3.11, Flask, SQLite, server-rendered HTML; coding standard PEP 8.
- **Process:** incremental model - Increment 1 delivers all "Must have" features (Section 4) by the November 2026 evaluation.

2.6 User Documentation

- README with installation, demo accounts and run instructions (delivered).
- In-app guidance: field placeholders, plain-language error messages and Trust Score breakdown (delivered).
- Test report with executed test cases - Appendix E (delivered).
- One-page vendor quick-start guide in Hindi and English (planned, next increment).

2.7 Assumptions and Dependencies

#	Assumption / dependency
A-1	Loan tiers follow the restructured PM SVANidhi scheme (Aug 2025): ₹15,000 / 12 months, ₹25,000 / 18 months, ₹50,000 / 36 months. The original 2020 scheme used ₹10,000 / ₹20,000 / ₹50,000; tier values are configurable in one place (TIERS in loan.py).
A-2	An interest rate of 12% p.a. (reducing balance) is assumed for EMI calculation until the partner lender confirms its rate (TBD-2). Government interest subsidy is credited by the lender, outside this system.
A-3	Vendors own or can borrow a smartphone, or are assisted at a Common Service Centre.
A-4	Self-reported sales are honest enough to be useful; later increments will cross-check against UPI receipts to limit inflation.
A-5	Trust Score weights (40/40/10/10) are initial values to be re-calibrated with real repayment data (TBD-5).
D-1	Availability of a UPI payment gateway (TBD-3) and an SMS/OTP provider (TBD-4).
D-2	A partner bank/NBFC willing to accept the Trust Score as an input to its credit decision.
D-3	Municipal bodies providing vending-zone data and licence rules (TBD-1).

3. External Interface Requirements

3.1 User Interfaces

All screens share a top bar (logo, user name and role, Dashboard and Logout links) and show success or error messages in a coloured banner at the top of the page (green for success, red for errors).

Screen	Used by	Main elements
Landing page	Everyone	Short explanation of the 3-step journey; Register and Log in buttons
Register / Login	Everyone	Name, mobile number, role (picked from four cards), password (+confirm); demo accounts that fill the login form in one click
Vendor dashboard	Vendor	Summary cards (Trust Score, 30-day sales, active days, eligible limit / next EMI); Trust Score ring with risk band and a bar per component; 14-day sales chart with the daily-sale form; micro-credit card showing the three tiers, the apply form and the loan list; business profile form
Loan detail	Vendor, Officer	Loan summary; EMI schedule table; Pay button on the next due installment
Officer dashboard	Bank officer	Count cards per status; applications table (pending first) with a colour-coded Trust Score; Approve/Reject with remarks; Disburse; Run default check
Find vendors	Customer	Search by area; vendor cards with business type, zone and UPI ID
Municipal dashboard	Municipal officer	Vendor registry by zone (licence screens in next increment)

- Layout is responsive (single column below 640 px) and uses large touch targets for vendors.
- Loan status is shown as colour-coded pills (Pending, Approved, Disbursed, Rejected, Defaulted, Closed).
- Error messages state what is wrong and how to fix it, e.g. "Amount exceeds Tier 1 limit of Rs 15,000".

3.2 Hardware Interfaces

The system has no direct hardware interface. It runs in a browser on smartphones and PCs. Later increments may use the phone's GPS (vendor location) and camera (scanning a vendor's UPI QR code) through standard browser APIs.

3.3 Software Interfaces

Component	Version	Purpose / data exchanged
Python	3.11	Runtime for the application server
Flask (with Werkzeug, Jinja2)	3.1	Web framework, routing, sessions, HTML templates, password hashing
SQLite / PostgreSQL	3.x / 15	Persistent storage of all entities (Section 6.1)
UPI payment gateway	TBD-3	Out: payment request (amount, payee UPI ID). In: payment confirmation / failure
SMS / OTP gateway	TBD-4	Out: OTP, EMI reminders, loan status alerts
Map service (OpenStreetMap / Leaflet)	Future	Display vending zones and nearby vendors
Lender core banking system	Future (REST/JSON)	Out: approved loan details. In: disbursal and repayment confirmation

3.4 Communications Interfaces

- All client-server traffic uses HTTPS (TLS 1.2 or higher) in deployment; forms are submitted with HTTP POST.
- Sessions use a signed, HttpOnly cookie; no sensitive data is placed in URLs.
- Notifications (EMI reminders, loan decisions) will be sent by SMS in the next increment.
- Integration with lenders will use a JSON REST API over HTTPS with API-key authentication (future).

4. System Features

Functional requirements are organised by feature. Priorities use the MoSCoW technique:

Priority	Features	Release
Must have	4.1 Registration & Authentication; 4.2 Vendor Profile & Daily Sales; 4.3 Trust Scoring Engine; 4.4 Loan Application & Progressive Lending; 4.5 Loan Review & Disbursal; 4.6 EMI Repayment	1.0 (implemented)
Should have	4.7 Vending Zone Licensing; 4.9 Customer: Locate & Pay Vendors	1.1
Could have	4.8 Dispute / Eviction Case Logging; 4.10 Peer-Guarantor Group Lending; vendor ratings; vending-zone map	1.2
Won't have (this release)	Aadhaar e-KYC, direct credit-bureau reporting, real fund transfer, voice interface	Later

4.1 User Registration & Authentication [Implemented]

4.1.1 Description and Priority

Lets every user class create an account and log in securely. Priority: **Must have / High**.

4.1.2 Stimulus/Response Sequences

Stimulus (user action)	System response
User opens Register, fills name, mobile, role, password	Validates input, stores the account with a hashed password, shows "Registration successful"
User enters mobile number and password on Login	Verifies credentials and opens the role's dashboard
User enters a wrong password	Shows "Invalid mobile number or password"; counts the failed attempt
Fifth consecutive wrong password	Locks the account and shows the lock message
User opens a page of another role	Redirects to own dashboard with "You are not allowed to open that page"

4.1.3 Functional Requirements

ID	Requirement
FR-AUTH-1	The system shall let a new user register with full name, mobile number, password and one role: Vendor, Bank/NBFC Officer, Municipal Officer or Customer. The name shall have at most 50 characters and contain only letters, spaces, dots and apostrophes.
FR-AUTH-2	The mobile number shall be a valid 10-digit Indian number starting with 6-9 and shall be unique; otherwise the system shall show "Enter a valid 10-digit mobile number" or "Mobile number already registered".
FR-AUTH-3	The password shall have at least 6 characters including at least one letter and one digit.
FR-AUTH-4	Passwords shall be stored only as salted one-way hashes.
FR-AUTH-5	On successful login the system shall start a session and open the dashboard of the user's role.

ID	Requirement
FR-AUTH-6	After 5 consecutive failed login attempts the account shall be locked; a successful login resets the counter.
FR-AUTH-7	Every page except Landing, Register and Login shall require a logged-in user, and role-specific pages shall be accessible only to that role.
FR-AUTH-8	Mobile-number verification by OTP shall be added when the SMS gateway is available (TBD-4).

4.2 Vendor Profile & Daily Sales Logging [Implemented]

4.2.1 Description and Priority

Creates the vendor's digital business identity and the sales history that feeds the Trust Score.
Priority: **Must have / High**.

4.2.2 Stimulus/Response Sequences

Stimulus (user action)	System response
Vendor fills business type, vending zone, years of vending, UPI ID and saves	Saves or updates the profile; Trust Score card appears
Vendor enters date, amount, payment mode and clicks Log sale	Validates and stores the entry, refreshes the 30-day statistics and Trust Score
Vendor logs a second entry for the same date	Rejects with "Sales already logged for this date"

4.2.3 Functional Requirements

ID	Requirement
FR-VEN-1	A vendor shall be able to create and update a profile: business type, vending zone, years of vending and optional UPI ID.
FR-VEN-2	Business type and vending zone are mandatory; years of vending shall be 0-60.
FR-VEN-3	A vendor shall be able to log one sales entry per day with date, amount and payment mode (Cash or UPI).
FR-VEN-4	The amount shall be a valid number greater than 0 and not more than ₹1,00,000 per day.
FR-VEN-5	The sale date shall not be in the future nor more than 90 days old, and only one entry per date is allowed.
FR-VEN-6	Sales can be logged only after the profile is complete.
FR-VEN-7	The dashboard shall show active days, total and daily average sales for the last 30 days and the 10 most recent entries.

4.3 Alternative Credit (Trust) Scoring Engine [Implemented]

4.3.1 Description and Priority

Replaces the missing CIBIL score with a transparent score built from data the platform observes.
Priority: **Must have / High**. Detailed processing is shown in the Level-2 DFD (Figure 8).

4.3.2 Stimulus/Response Sequences

Stimulus (user action)	System response
Vendor opens the dashboard	Computes and shows the score, risk band and the four component points
Vendor applies for a loan	Computes the score; blocks the application if the score is below 40

4.3.3 Functional Requirements

ID	Requirement
FR-TS-1	The system shall compute a Trust Score from 0 to 100 from the vendor's profile, last-30-day sales, EMI history and vouches whenever the dashboard is opened or a loan is applied for.
FR-TS-2	Score = Sales consistency (active days / 30 × 40) + Repayment history (on-time EMIs / EMIs due × 40) + Peer vouching (min(vouches, 3) / 3 × 10) + Vending tenure (min(years, 5) / 5 × 10), rounded to the nearest integer.
FR-TS-3	A vendor with no EMI due yet shall receive a neutral 20 repayment points.
FR-TS-4	Risk band: 70-100 LOW RISK, 40-69 MEDIUM RISK, 0-39 HIGH RISK.
FR-TS-5	A score of at least 40 is required to apply for a loan.
FR-TS-6	The vendor shall see the points earned in each component, so the score is explainable.

4.4 Loan Application & Progressive Lending [Implemented]

4.4.1 Description and Priority

Lets an eligible vendor apply for a working-capital loan whose limit grows as earlier loans are repaid. Priority: **Must have / High**.

4.4.2 Stimulus/Response Sequences

Stimulus (user action)	System response
Vendor enters an amount within the shown tier and clicks Apply	Checks eligibility, stores the application as PENDING with a snapshot of the Trust Score
Vendor enters an amount above the tier limit	Rejects with "Amount exceeds Tier n limit of Rs X"
Vendor with a PENDING application clicks Withdraw	Marks the application WITHDRAWN

4.4.3 Functional Requirements

ID	Requirement
FR-LN-1	An eligible vendor shall be able to apply for a loan by entering the amount.
FR-LN-2	Tier limits: Tier 1 up to ₹15,000 for 12 months; Tier 2 up to ₹25,000 for 18 months; Tier 3 up to ₹50,000 for 36 months.
FR-LN-3	The eligible tier shall be (number of fully repaid loans + 1), at most 3.
FR-LN-4	The amount shall be a valid number greater than 0 and not more than the tier limit.
FR-LN-5	A vendor may have only one active loan (Pending, Approved, Disbursed or Defaulted) at a time.
FR-LN-6	The application shall store the tier, tenure, interest rate and Trust Score at the time of applying, with status PENDING; the vendor may withdraw it while it is PENDING.

4.5 Loan Review & Disbursal (Bank / NBFC) [Implemented]

4.5.1 Description and Priority

Gives the lender a queue of applications with the Trust Score and controls the loan life cycle. Priority: **Must have / High**. Life cycle: Figure 5.

4.5.2 Stimulus/Response Sequences

Stimulus (user action)	System response
Officer opens the dashboard	Lists applications, PENDING first, with vendor, amount, tier and Trust Score
Officer clicks Approve	Sets status APPROVED
Officer clicks Reject without remarks	Rejects the action: "Remarks are required when rejecting"
Officer clicks Disburse on an APPROVED loan	Sets DISBURSED and generates the EMI schedule

Stimulus (user action)	System response
Officer runs the default check	Marks loans with an EMI more than 90 days overdue as DEFAULTED

4.5.3 Functional Requirements

ID	Requirement
FR-BNK-1	The officer shall see all applications with vendor name, mobile, date, tier, amount, Trust Score and status, with PENDING applications first.
FR-BNK-2	Only PENDING applications can be approved or rejected.
FR-BNK-3	Rejection shall require remarks, which are shown to the vendor.
FR-BNK-4	Only users with the Bank/NBFC Officer role can review or disburse loans.
FR-BNK-5	Only APPROVED loans can be disbursed; disbursal sets the status to DISBURSED and records the date.
FR-BNK-6	On disbursal the system shall generate a monthly reducing-balance EMI schedule: EMI = P·r·(1+r)^n / ((1+r)^n − 1), r = annual rate / 12; the last EMI absorbs rounding.
FR-BNK-7	A DISBURSED loan with any EMI unpaid more than 90 days after its due date shall be marked DEFAULTED.

4.6 EMI Repayment Tracking [Implemented]

4.6.1 Description and Priority

Records repayments, which close the loan, unlock the next tier and raise the Trust Score. Priority: **Must have / High**.

4.6.2 Stimulus/Response Sequences

Stimulus (user action)	System response
Vendor opens a loan	Shows the schedule with due dates, EMI split and status
Vendor pays the next due EMI	Marks it PAID (on time or late) and refreshes the schedule
Vendor pays the last EMI	Marks the loan CLOSED and shows that the next tier is unlocked

4.6.3 Functional Requirements

ID	Requirement
FR-EMI-1	The vendor shall see each installment's number, due date, EMI, principal, interest and status.
FR-EMI-2	Installments shall be paid in order; paying a later one first shows "Pay earlier installment #n first".
FR-EMI-3	The amount paid shall not be less than the EMI due.
FR-EMI-4	An installment that is already paid cannot be paid again.
FR-EMI-5	The system shall record the payment date and whether it was on or before the due date; this feeds the repayment component of the Trust Score.
FR-EMI-6	When all installments are paid the loan shall become CLOSED and the vendor becomes eligible for the next tier.

4.7 Vending Zone Licensing

4.7.1 Description and Priority

Digitises the vending certificate process of the Street Vendors Act 2014 (R5). Priority: **Should have / Medium**.

4.7.2 Stimulus/Response Sequences

Stimulus (user action)	System response
Vendor submits a licence application for a zone	Creates the application and notifies the ward's municipal officer
Municipal officer approves	Issues a licence number with issue and expiry dates
Licence is 30 days from expiry	Sends a renewal reminder to the vendor

4.7.3 Functional Requirements

ID	Requirement
FR-LIC-1	A vendor shall be able to apply for a vending licence for a chosen vending zone.
FR-LIC-2	The municipal officer of that ward shall approve or reject the application with remarks.
FR-LIC-3	An approved licence shall have a unique number, issue date and expiry date (validity TBD-1).
FR-LIC-4	The vendor shall be reminded 30 days before expiry and be able to renew online.
FR-LIC-5	Vendors shall see a map of permitted vending zones.

4.8 Dispute / Eviction Case Logging

4.8.1 Description and Priority

Gives municipal officers a record of disputes and eviction notices. Priority: **Could have / Low**.

4.8.2 Stimulus/Response Sequences

Stimulus (user action)	System response
Officer logs a case against a vendor or zone	Creates the case with status OPEN and notifies the vendor
Officer resolves the case	Records the resolution and closes the case

4.8.3 Functional Requirements

ID	Requirement
FR-CASE-1	A municipal officer shall log a case with type (dispute, eviction notice, relocation), description, vendor and date.
FR-CASE-2	The affected vendor shall be notified and able to view the case.
FR-CASE-3	A case shall move OPEN → UNDER REVIEW → RESOLVED, with remarks at each step.
FR-CASE-4	Eviction cases shall reference the vendor's licence where one exists.

4.9 Customer: Locate, Pay & Rate Vendors

4.9.1 Description and Priority

Brings customers onto the platform and gives vendors a source of digital (verifiable) sales. Priority: **Should have / Medium** (vendor search by area is already implemented).

4.9.2 Stimulus/Response Sequences

Stimulus (user action)	System response
Customer searches by area	Lists matching vendors with business type, zone and UPI ID
Customer pays a vendor	Initiates a UPI payment and stores the receipt
Customer rates a vendor after paying	Stores hygiene and service ratings

4.9.3 Functional Requirements

ID	Requirement

ID	Requirement
FR-CUS-1	A customer shall be able to search vendors by area or vending zone. [Implemented]
FR-CUS-2	A customer shall be able to view nearby vendors on a map (future).
FR-CUS-3	A customer shall be able to pay a vendor through UPI; the payment shall also be recorded as a verified sale for the vendor.
FR-CUS-4	After a payment, a customer may rate hygiene and service from 1 to 5 stars with a comment.

4.10 Peer-Guarantor Group Lending

4.10.1 Description and Priority

Uses the joint-liability model common in Indian microfinance. Priority: **Could have / Low**.

4.10.2 Stimulus/Response Sequences

Stimulus (user action)	System response
Vendor A vouches for vendor B	Records the vouch; B's peer-vouching points increase
Vendors form a lending circle	Creates a group whose members see each other's repayment status

4.10.3 Functional Requirements

ID	Requirement
FR-PEER-1	A vendor shall be able to vouch for another vendor once; a vendor cannot vouch for themself.
FR-PEER-2	At most 3 vouches count toward the Trust Score (FR-TS-2).
FR-PEER-3	Vendors shall be able to form lending circles of 3-5 members (rules TBD-7).
FR-PEER-4	If a member's EMI is overdue, the circle's members shall be notified.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement
NFR-PERF-1	95% of page requests shall complete within 2 seconds with 100 concurrent users.
NFR-PERF-2	Trust Score computation for one vendor shall take less than 0.5 seconds.
NFR-PERF-3	Each page shall be under 500 KB so it loads within 5 seconds on a 3G connection.
NFR-PERF-4	The system shall support at least 10,000 registered vendors and 1 million sales records without redesign.

5.2 Safety Requirements

ID	Requirement
NFR-SAFE-1	No loan shall be disbursed automatically; a human bank officer must approve and disburse (human-in-the-loop). The Trust Score is an input to, not a replacement for, the lender's decision.
NFR-SAFE-2	Every approval, rejection and disbursal shall record the officer and remarks for audit.
NFR-SAFE-3	Database backups shall be taken daily and kept for 30 days so financial records are not lost.
NFR-SAFE-4	Invalid input shall never corrupt stored data; each operation is validated before it is saved.

5.3 Security Requirements

ID	Requirement
NFR-SEC-1	Passwords shall be stored as salted hashes (Werkzeug scrypt/PBKDF2); plain text is never stored.
NFR-SEC-2	Accounts shall lock after 5 consecutive failed logins (FR-AUTH-6).
NFR-SEC-3	Role-based access control shall be enforced on the server for every request (FR-AUTH-7).
NFR-SEC-4	All database queries shall use parameterised statements to prevent SQL injection.
NFR-SEC-5	Templates shall auto-escape user input to prevent cross-site scripting.
NFR-SEC-6	Production deployment shall use HTTPS and a secret key supplied through an environment variable.
NFR-SEC-7	A vendor shall only see their own sales and loans; officers see only data needed for their role.
NFR-SEC-8	Personal data handling shall comply with the DPDP Act 2023: consent at registration, purpose limitation and the right to request deletion.

5.4 Software Quality Attributes

ID	Requirement
NFR-USE-1	A first-time vendor shall be able to log a day's sale in under 30 seconds.
NFR-USE-2	Every error message shall say what went wrong and how to correct it.
NFR-REL-1	Loan and EMI state changes shall be atomic (committed together or not at all).
NFR-AVL-1	The service shall be available 99% of the time, excluding announced maintenance.
NFR-MNT-1	Business logic shall be separated from the web layer (modules package) and follow PEP 8.
NFR-MNT-2	Loan tiers, interest rate and score weights shall be defined as constants in one place each.
NFR-TST-1	Every Must-have feature shall have automated tests; all tests shall pass before a release.
NFR-PRT-1	The application shall run unchanged on Windows and Linux and in any modern browser.

5.5 Business Rules

ID	Rule
BR-1	Only a Bank/NBFC Officer may approve, reject, disburse or mark a loan as defaulted.
BR-2	A vendor may hold only one active loan at a time.
BR-3	Loan limits grow progressively: a tier is unlocked only after the previous loan is fully repaid.
BR-4	A vendor needs a Trust Score of at least 40 to apply.
BR-5	A rejected application must carry the officer's reason.
BR-6	An EMI unpaid for more than 90 days after its due date makes the loan DEFAULTED.
BR-7	A municipal officer may act only on licences and cases in their own ward.
BR-8	A vendor cannot vouch for themself, and only 3 vouches count toward the score.

6. Other Requirements

6.1 Database Requirements

Table	Key columns	Notes
users	user_id, name, phone (unique), password_hash, role, failed_attempts, is_locked	All user classes
vendor_profiles	vendor_id (FK users), business_type, vending_zone, years_vending, upi_id	1 per vendor
sales	sale_id, vendor_id, sale_date, amount, payment_mode	Unique (vendor_id, sale_date)
vouches	vouch_id, voucher_id, vendor_id	Unique (voucher_id, vendor_id)
loans	loan_id, vendor_id, tier, amount, tenure_months,	Status per Figure 5

Table	Key columns	Notes
	interest_rate, trust_score, status, officer_id, remarks, applied_on, disbursed_on	
emi_installments	emi_id, loan_id, installment_no, due_date, emi_amount, principal_part, interest_part, status, paid_on, paid_on_time	Unique (loan_id, installment_no)
licences, cases, ratings	(next increment)	Classes VendingLicence, DisputeCase, Rating

6.2 Legal and Regulatory Requirements

- Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act, 2014 - licensing and Town Vending Committees.
- RBI Guidelines on Digital Lending (2022) - disbursal and repayment directly between borrower and regulated lender; loan terms shown to the borrower before acceptance.
- Digital Personal Data Protection Act, 2023 - consent, purpose limitation, data minimisation.

6.3 Internationalisation Requirements

The interface shall support English and Hindi in release 1.1, with further regional languages added through translation files (TBD-6). Amounts shall use Indian digit grouping (₹1,00,000).

6.4 Reuse Objectives

The modules package (auth, vendor, credit_score, loan) has no web dependencies, so the same code can serve a future mobile app or lender API.

Appendix A: Glossary

Term	Meaning
CIBIL	TransUnion CIBIL - India's best-known credit bureau; issues credit scores
DFD	Data Flow Diagram - shows how data moves between processes, data stores and external entities
DPDP Act	Digital Personal Data Protection Act, 2023
EMI	Equated Monthly Installment - fixed monthly repayment covering principal and interest
KYC	Know Your Customer - identity verification
MoSCoW	Prioritisation technique: Must have, Should have, Could have, Won't have
NBFC	Non-Banking Financial Company, regulated by the RBI
PM SVANidhi	PM Street Vendor's AtmaNirbhar Nidhi - GoI micro-credit scheme for street vendors
Peer Guarantor	A vendor who vouches for another vendor's reliability
Progressive lending	Loan limits that increase after each successful repayment
Tier	Loan level (1-3) that sets the maximum amount and tenure
Trust Score	VendorSetu's 0-100 alternative credit score
TVC / ULB	Town Vending Committee / Urban Local Body (municipal corporation or council)
UML	Unified Modeling Language
UPI	Unified Payments Interface - India's real-time payment system

Appendix B: Analysis Models

This appendix contains the UML and structured-analysis models of VendorSetu: Use Case, Class, Sequence and State Transition diagrams, and Data Flow Diagrams at Levels 0, 1 and 2.

B.1 Use Case Diagram

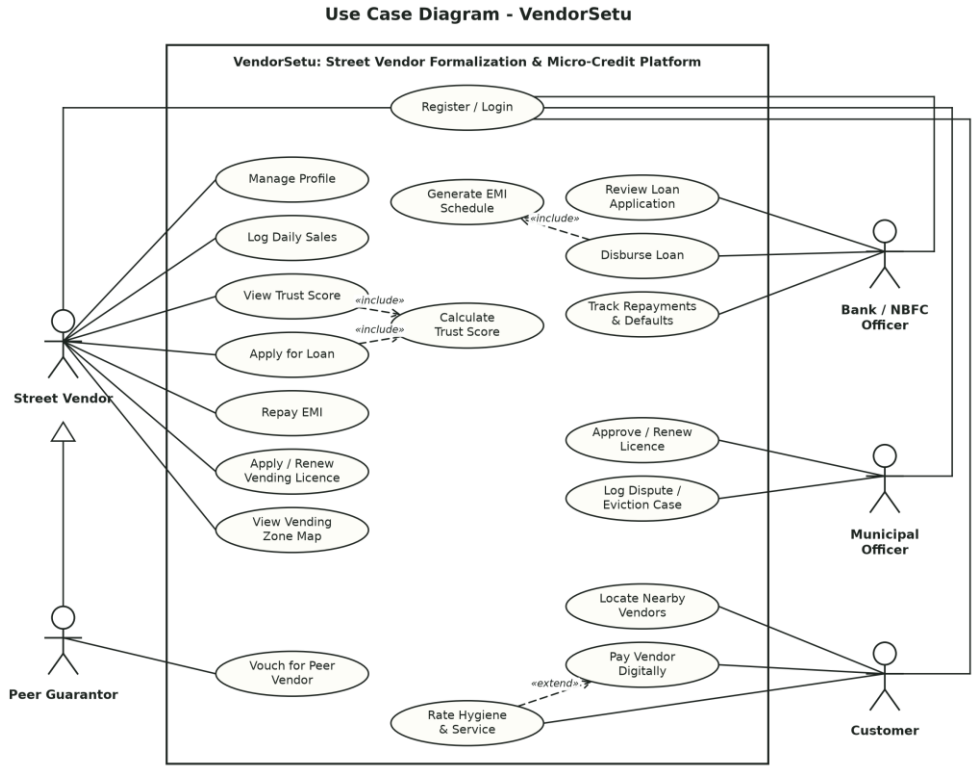


Figure 1: Use Case Diagram

How to read it: stick figures are actors (people or systems outside VendorSetu); ovals are use cases (goals an actor achieves); the rectangle is the system boundary. A solid line means the actor takes part in the use case. **«include»** (dashed arrow) means the base use case always performs the included one - applying for a loan always

calculates the Trust Score. **«extend»** means optional behaviour - a customer may rate a vendor after paying. The hollow triangle shows that a Peer Guarantor is a specialised Street Vendor.

Use case descriptions

Field	UC-05 Apply for Loan	UC-03 Log Daily Sales
Primary actor	Street Vendor	Street Vendor
Pre-conditions	Logged in; profile complete; no active loan	Logged in; profile complete
Main flow	1. Vendor opens dashboard and sees eligible tier. 2. Enters amount and clicks Apply. 3. System checks for an active loan. 4. System calculates the Trust Score («include»). 5. System checks score ≥ 40 and amount ≤ tier limit. 6. System saves the application as PENDING and confirms.	1. Vendor enters date, amount and payment mode. 2. Clicks Log sale. 3. System validates the entry. 4. System saves it and refreshes the 30-day summary and Trust Score.
Alternate flows	3a. Active loan exists → error, end. 5a. Score < 40 → "not eligible" with score. 5b. Amount > limit → error with the limit.	3a. Amount ≤ 0 or > ₹1,00,000 → error. 3b. Future date → error. 3c. Date already logged → error.
Post-conditions	Loan record with status PENDING and Trust Score snapshot	One new SALES record
Requirements	FR-LN-1 to FR-LN-6, FR-TS-5	FR-VEN-3 to FR-VEN-7

B.2 Class Diagram

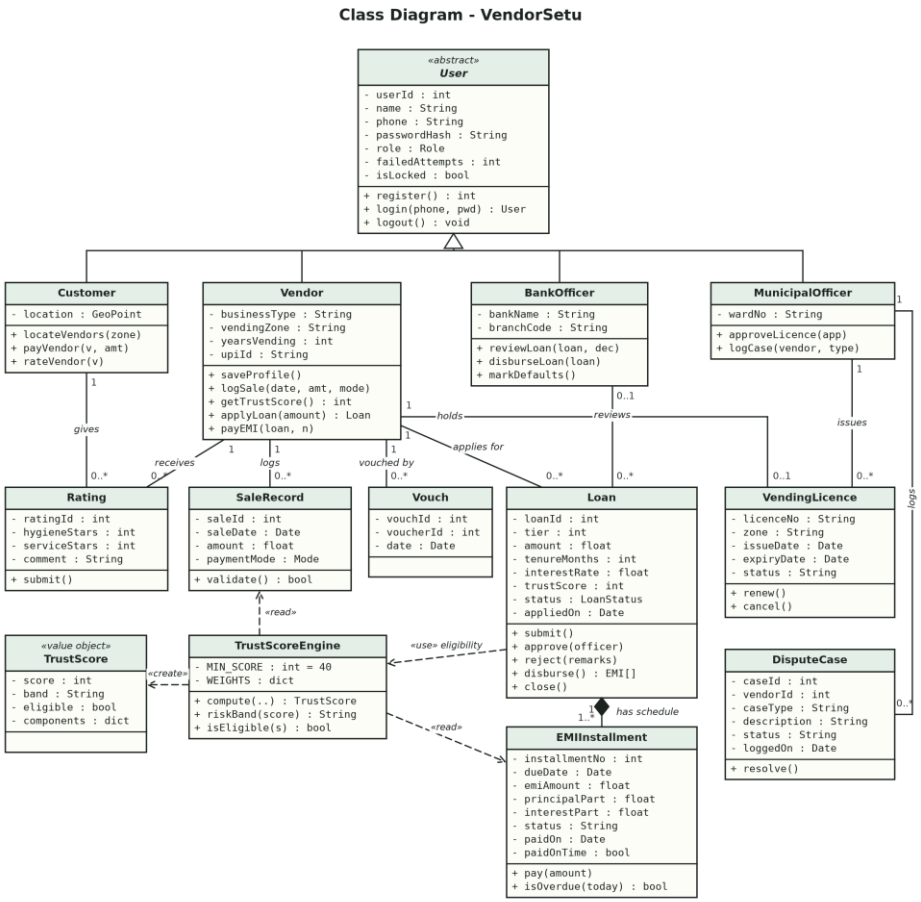


Figure 2: Class Diagram

How to read it: each box is a class with three parts - name, attributes (- private) and operations (+ public). The hollow-triangle tree shows **inheritance**: Customer, Vendor, BankOfficer and MunicipalOfficer are kinds of the abstract User. Plain lines are **associations** with multiplicities (1, 0..1, 0..*), e.g. one Vendor applies for zero or more Loans. The filled diamond is **composition**: EMIInstallments cannot exist without their Loan. Dashed arrows are **dependencies** - TrustScoreEngine reads SaleRecords and EMIInstallments and creates a TrustScore value object, and Loan uses the engine to check eligibility.

B.3 Sequence Diagrams

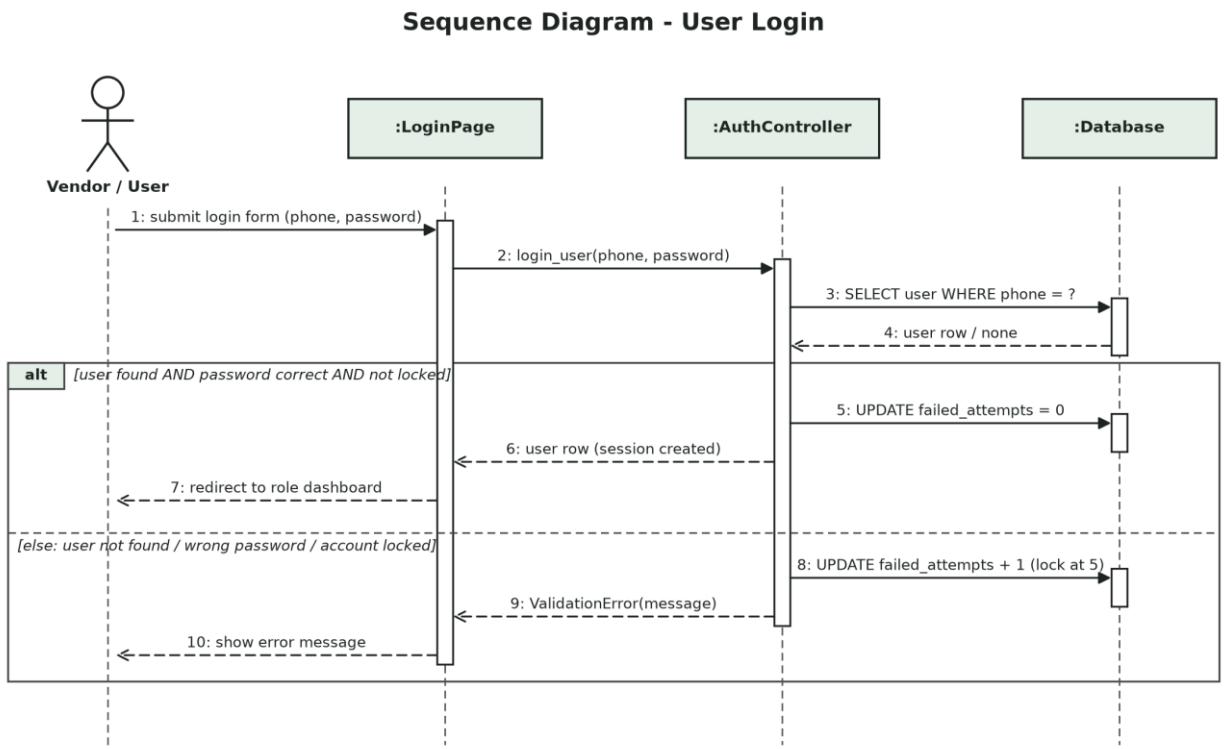


Figure 3: Sequence Diagram - User Login

How to read it: participants are along the top, time flows downward on the dashed lifelines, and the thin rectangles show when an object is active. Solid arrows are calls, dashed arrows are returns. The **alt** frame shows the two possible outcomes: a valid login opens the dashboard; an invalid one increments the failed-attempt counter (locking the account at 5) and shows an error.

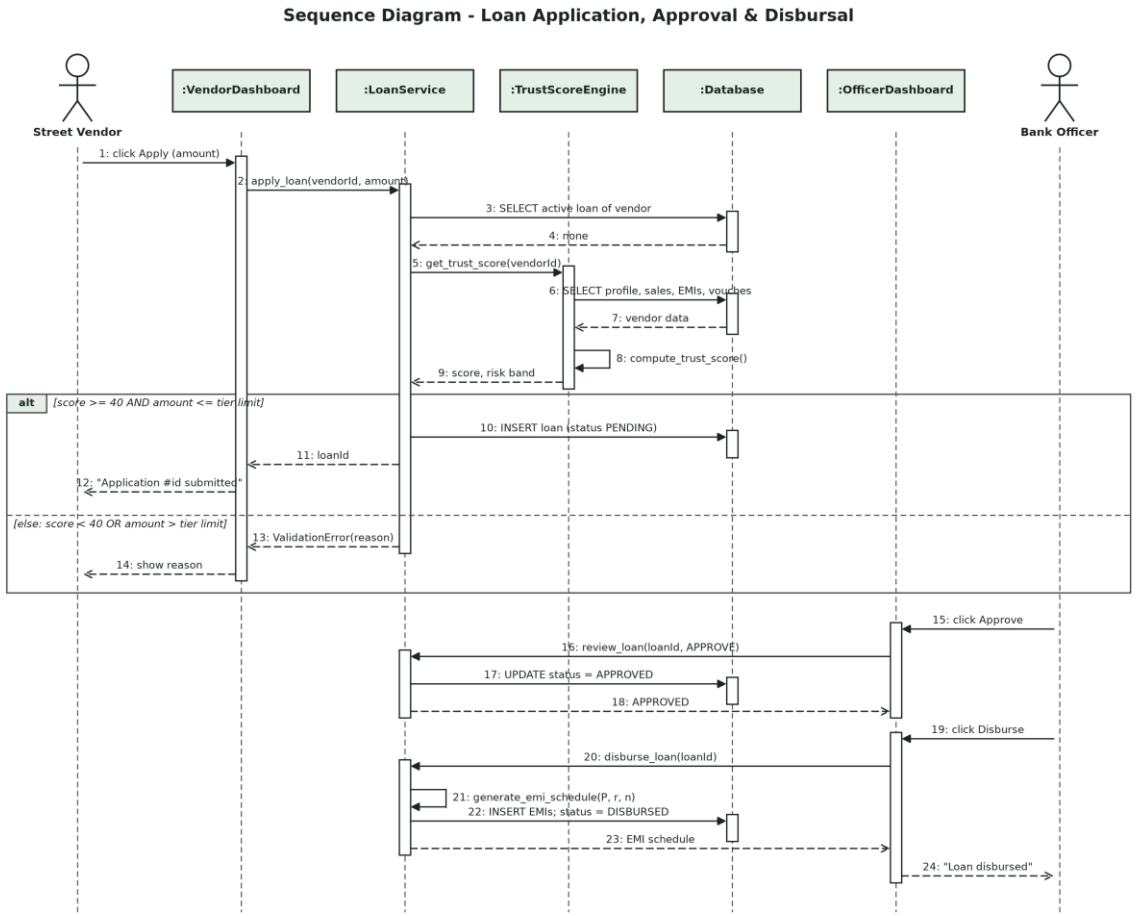


Figure 4: Sequence Diagram - Loan Application, Approval and Disbursal

Messages 1-14 show the vendor's application: the LoanService checks for an active loan, asks the TrustScoreEngine for a score (which reads the database and computes the score itself), and then either saves the PENDING application or returns the reason for refusal. Messages 15-24 show the bank officer approving the loan and disbursing it, which generates and saves the EMI schedule.

Message names are the real function names in the code (login_user in modules/auth.py; apply_loan, review_loan, disburse_loan, generate_emi_schedule in modules/loan.py; get_trust_score, compute_trust_score in modules/credit_score.py). Pages (:LoginPage, :VendorDashboard, :OfficerDashboard) are boundary objects served by app.py; :AuthController, :LoanService and :TrustScoreEngine are the control objects implemented by those modules; the entity classes they read and write are the ones in the Class Diagram (Figure 2).

B.4 State Transition Diagram

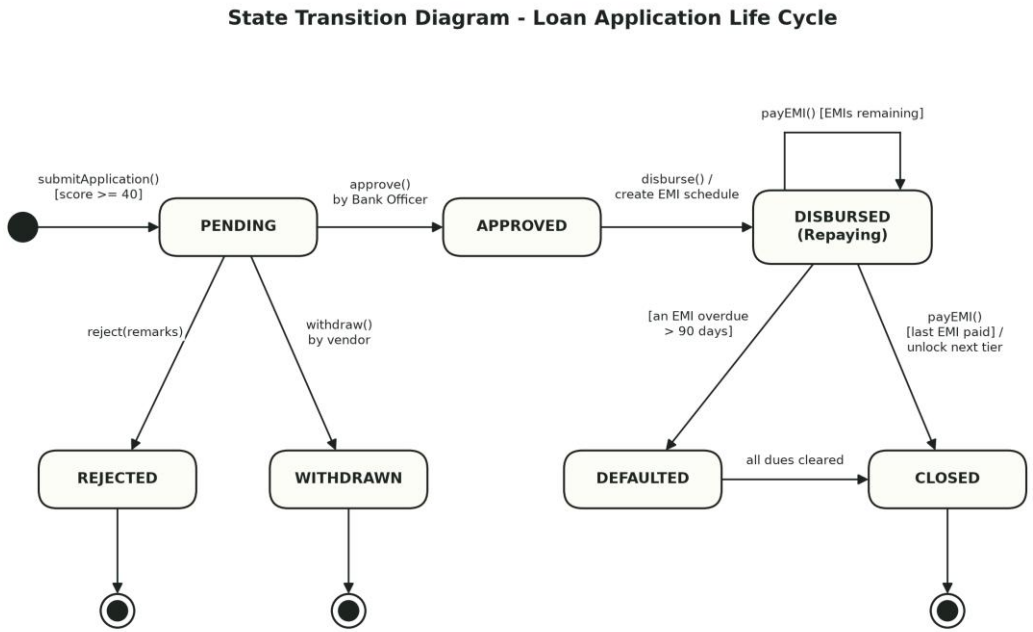


Figure 5: State Transition Diagram - Loan Application Life Cycle

State	Meaning	Leaves the state when
PENDING	Application submitted, waiting for the bank	Officer approves/rejects, or vendor withdraws
APPROVED	Bank has sanctioned the loan	Officer disburses (EMI schedule created)
DISBURSED	Money released; vendor is repaying	Last EMI paid (→ CLOSED) or an EMI is >90 days overdue (→ DEFAULTED)
DEFAULTED	Serious delay in repayment	All dues cleared (→ CLOSED)
REJECTED / WITHDRAWN / CLOSED	Final states	-

B.5 Data Flow Diagrams

Notation (Yourdon-DeMarco): rectangles are external entities, circles are processes, open-ended boxes are data stores, and labelled arrows are data flows.

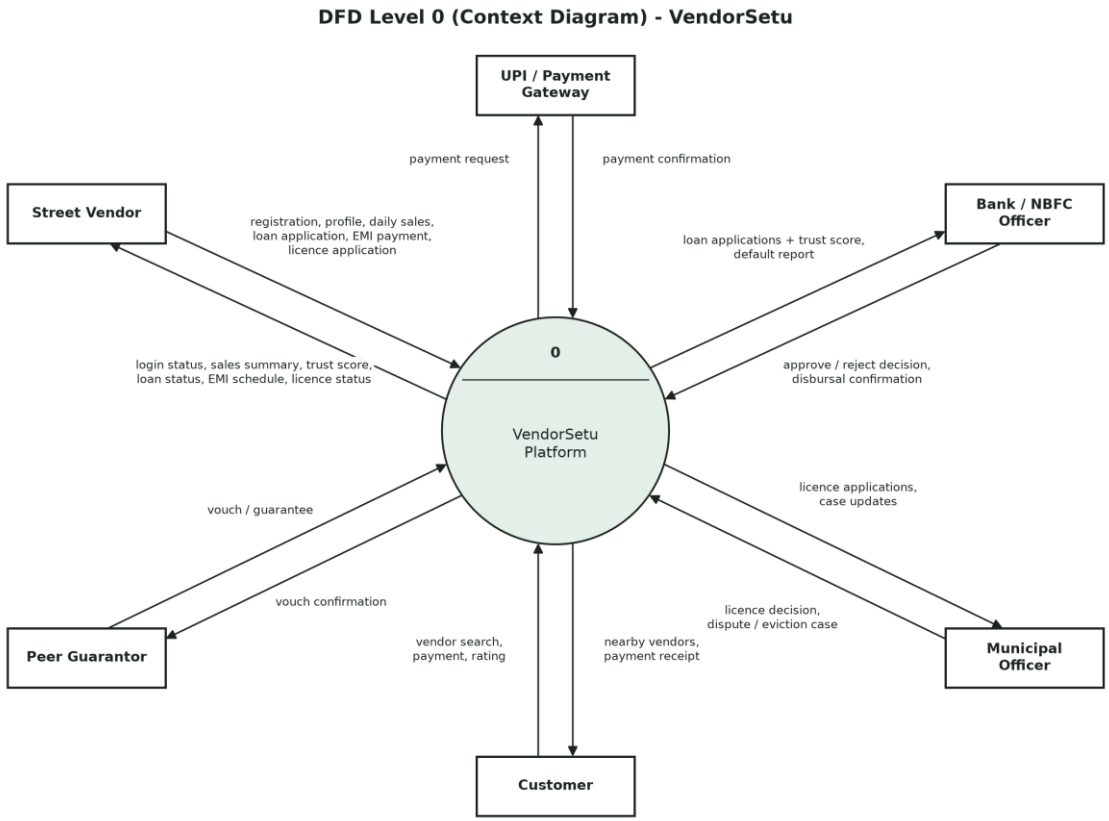


Figure 6: DFD Level 0 (Context Diagram)

The **context diagram** treats the whole of VendorSetu as one process (0) and shows only the data it exchanges with the six external entities.

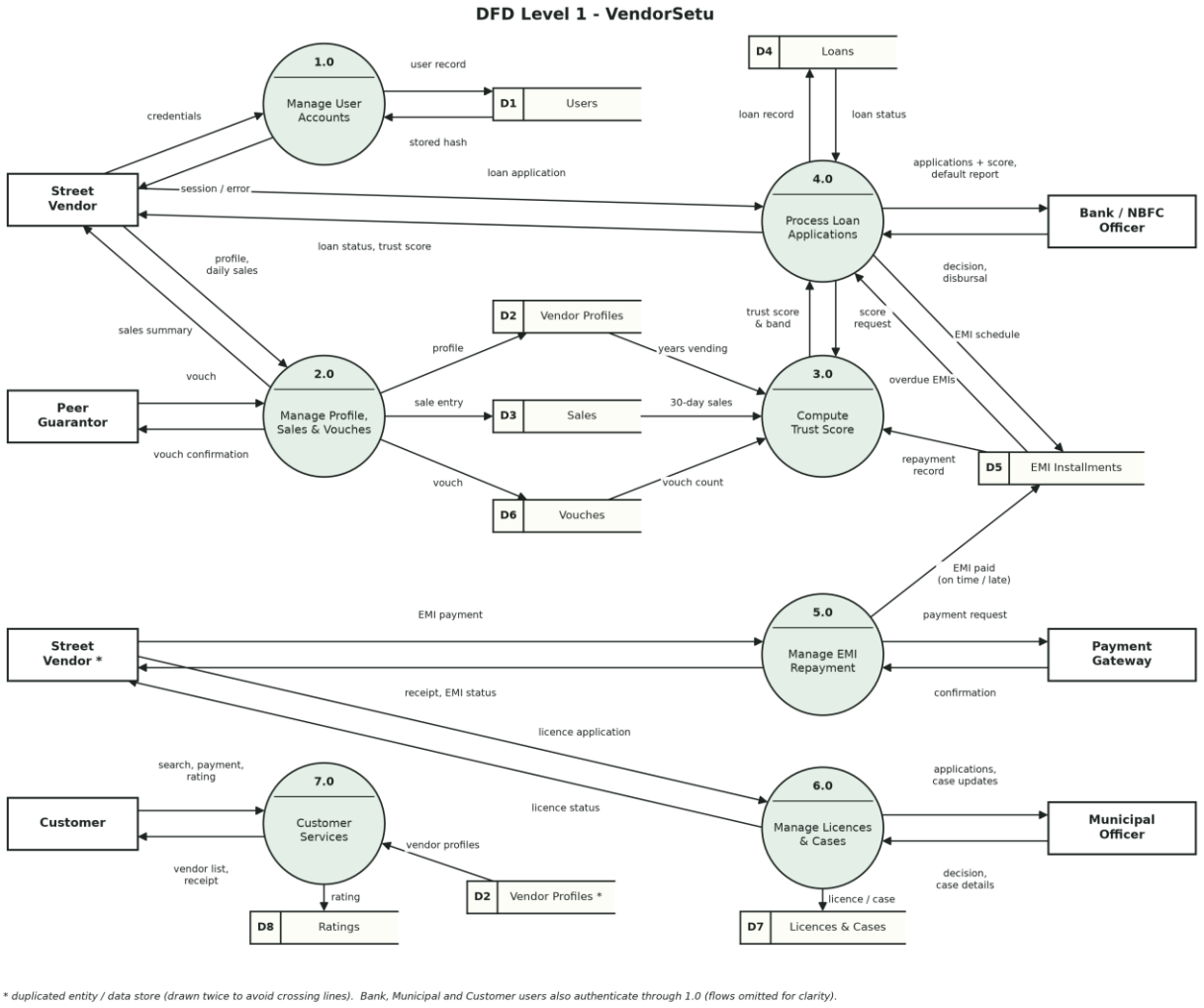


Figure 7: DFD Level 1

Level 1 breaks process 0 into seven processes and shows the data stores D1-D8. Process 3.0 (Trust Score) reads sales, profiles, vouches and EMI records and passes the score to 4.0 (Loans). Entities and stores marked * are drawn twice only to avoid crossing lines.

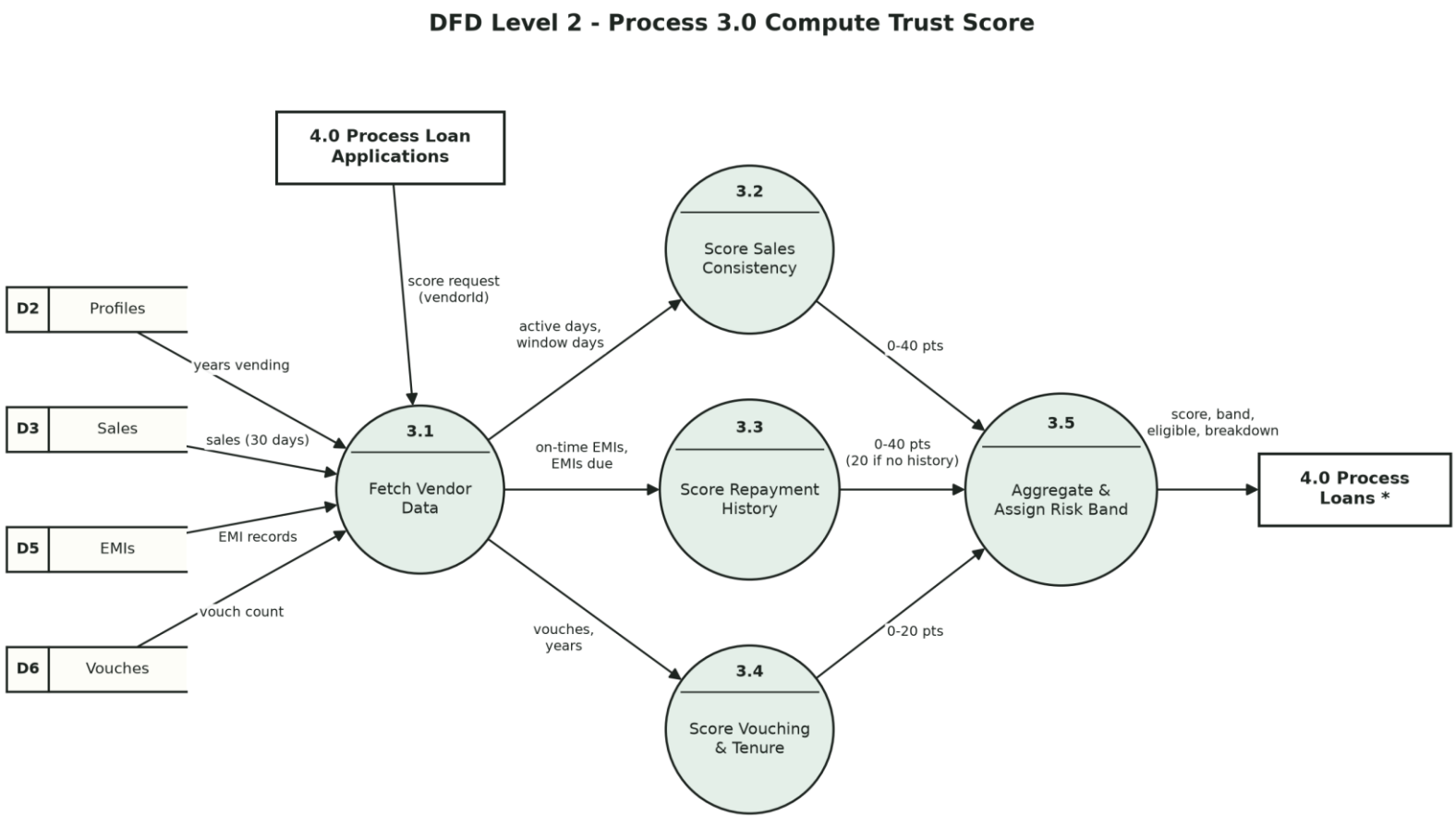


Figure 8: DFD Level 2 - Process 3.0 Compute Trust Score

Level 2 opens process 3.0: on a score request from 4.0, process 3.1 fetches the vendor's data, 3.2-3.4 compute the four score components, and 3.5 adds them (0-100), assigns the risk band and returns the result to 4.0, which shows it to the vendor. This matches compute_trust_score() in Appendix D.

Balancing: every data flow that enters or leaves a process at one level also appears at the next level - e.g. the vendor's "trust score" output in Level 0 is produced by 3.0 → 4.0 → Vendor in Level 1, and the "score request" from 4.0 into 3.0 in Level 1 is the input of process 3.1 in Level 2.

Appendix C: To Be Determined List

#	Item	Section
TBD-1	Licence validity period and fees - set by each city's Town Vending Committee	4.7
TBD-2	Partner lender's actual interest rate (12% p.a. assumed)	2.7, 4.5
TBD-3	UPI payment gateway provider	3.3
TBD-4	SMS/OTP provider for mobile verification and reminders	3.3, 4.1
TBD-5	Re-calibration of Trust Score weights using real repayment data	4.3
TBD-6	Regional languages beyond Hindi and English	6.3
TBD-7	Lending-circle size and joint-liability rules	4.10

Appendix D: Implementation (Release 1.0 Prototype)

D.1 Development Approach and Technology

The project follows an **incremental model**: Increment 1 implements the six "Must have" features end-to-end (four code modules plus the web interface); later increments add the Should/Could features. Technology: Python 3.11, Flask 3.1, SQLite, Jinja2 templates, pytest.

D.2 Coded Modules

Module	Source file	Features / requirements	Key functions
1. Registration & Authentication	app/modules/auth.py	4.1 - FR-AUTH-1..7	register_user, login_user
2. Vendor Profile & Daily Sales	app/modules/vendor.py	4.2 - FR-VEN-1..7	save_profile, log_daily_sale, sales_summary
3. Trust Scoring Engine	app/modules/credit_score.py	4.3 - FR-TS-1..6	compute_trust_score, get_trust_score
4. Loan Management (application, review, disbursal, EMI)	app/modules/loan.py	4.4-4.6 - FR-LN, FR-BNK, FR-EMI	apply_loan, review_loan, disburse_loan, generate_emi_schedule, pay_emi
Web interface	app/app.py, app/templates/	3.1 User Interfaces	Flask routes per role

D.3 Project Structure

```
SE_Project/
  app/
    app.py               Flask routes (web layer)
    database.py          SQLite schema + connection helper
    seed_demo.py         creates demo users and data
    modules/
      auth.py            Module 1 - registration & login
      vendor.py          Module 2 - profile & daily sales
      credit_score.py    Module 3 - Trust Score engine
      loan.py            Module 4 - loans, EMI schedule, repayment
    templates/, static/  HTML pages and stylesheet
  tests/
    test_cases.py        documented test cases (Appendix E)
    run_test_cases.py    executes cases, records actual output
    test_vendorsetu.py   pytest wrapper for the same cases
    test_integration.py  end-to-end tests through the web routes
```

```
  diagrams/              diagram source (Python) + PNG output
  docs/                  this SRS and its generator
```

D.4 Key Algorithms

Trust Score (FR-TS-2) - excerpt from app/modules/credit_score.py:

```
def compute_trust_score(active_days, window_days, on_time_emis, emis_due, vouches,
years_vending):
    """Pure calculation - returns a dict with the total score, risk band and each
component."""
    values = (active_days, window_days, on_time_emis, emis_due, vouches, years_vending)
    if any(v is None or v < 0 for v in values):
        raise ValidationError("Score inputs cannot be negative")
    if window_days == 0 or active_days > window_days:
        raise ValidationError("Active days cannot exceed the observation window")
    if on_time_emis > emis_due:
        raise ValidationError("On-time EMIs cannot exceed EMIs due")

    consistency = active_days / window_days * 40
    repayment = on_time_emis / emis_due * 40 if emis_due else NEUTRAL_REPAYMENT_POINTS
    peer = min(vouches, 3) / 3 * 10
    tenure = min(years_vending, 5) / 5 * 10

    score = round(consistency + repayment + peer + tenure)
    ...
```

EMI schedule (FR-BNK-6) - excerpt from app/modules/loan.py:

```
def generate_emi_schedule(principal, annual_rate, months, start_date):
    """Reducing-balance EMI schedule: EMI = P*r*(1+r)^n / ((1+r)^n - 1)."""
    if principal <= 0 or months <= 0:
        raise ValidationError("Principal and tenure must be positive")
    r = annual_rate / 12 / 100
    emi = principal / months if r == 0 else principal * r * (1 + r) ** months / ((1 + r) **
months - 1)
    emi = round(emi, 2)

    schedule, balance = [], principal
    for n in range(1, months + 1):
        interest = round(balance * r, 2)
        principal_part = round(emi - interest, 2)
        amount = emi
        if n == months:                      # last EMI clears any rounding difference
            principal_part = round(balance, 2)
            amount = round(principal_part + interest, 2)
        balance = round(balance - principal_part, 2)
        schedule.append({
            "installment_no": n,
            "due_date": add_months(start_date, n).isoformat(),
            "emi_amount": amount,
            "principal_part": principal_part,
            "interest_part": interest,
        })
    return schedule
```

D.5 How to Run

```
pip install flask pytest
cd app
python seed_demo.py        # creates demo data
python app.py              # open http://127.0.0.1:5000

Demo logins (mobile / password):
  Vendor        9876543210 / ramesh123
  Bank officer  9123456780 / bank123
  Customer      9988776655 / priya123
  Municipal     9090909090 / muni123

python -m pytest tests -v  # run all automated tests
```

D.6 Coding Standards Followed

- PEP 8 naming (snake_case functions, UPPER_CASE constants) and a module docstring naming the SRS section it implements.
- Business rules raise a single ValidationError type with a user-readable message; the web layer only displays it.
- All SQL uses ? parameters (no string concatenation of user input).
- Configuration values (tiers, interest rate, score weights, lock-out limit) are named constants.
- Functions accept an optional 'today' date so time-dependent rules can be tested deterministically.

Appendix E: Test Cases and Results

E.1 Test Strategy

- **System testing (80 cases):** each case is performed through the web interface exactly as a user would - fill a form, click a button - and the Expected / Actual Output is the message or value shown on the page, so every case can be repeated by hand in the demo.
- **Unit testing (12 cases):** the Trust Score formula is tested directly; the Expected Output shows the hand calculation (e.g. 40 + 20 + 10 + 10 = 80).
- **Design techniques:** positive testing; equivalence partitioning (every invalid class: empty, wrong format, non-numeric, nan / inf, out of range); boundary-value analysis on both sides of every limit (name 50 / 51, password 5 / 6, amount 1,00,000 / 1,00,000.01, date 90 / 91 days, score 39 / 40 and 69 / 70, tier limit 15,000 / 15,000.01, 90 / 91 days overdue); combination testing; security testing (role check, forged requests, SQL injection).
- **Combination testing:** 1,894 combinations of valid and invalid values in 7 forms (every value of every field with every value of the others) were submitted; 1,894 gave exactly the result required by the SRS, with no crash and no invalid data stored.
- **Defects found and fixed:** the first version of the code (commit 0ce1911) failed 9 of these test cases; 6 defects (e.g. 'nan' as an amount crashed the app, customers could open a vendor's loan page) were logged, fixed and re-tested - see E.8.
- **Integration testing:** two end-to-end scenarios (register → login → profile → sale → apply → approve → disburse → pay EMI; role-based access).
- **Test automation:** tests/run_test_cases.py executes every case on a fresh, empty database and copies the on-screen result into the Actual Output column; the same cases run under pytest.
- **Latest automated run:** pytest - 101 passed in 110.74s (0:01:50) (documented cases, combination suites and integration tests). The full step-by-step version of these cases, with a requirements traceability matrix, is in the separate Test Case Document (docs/Test_Cases_VendorSetu.docx).

E.2 User Registration & Login

Test Case #:	TC-AUTH-01 to TC-AUTH-20	**Test Case Name:**	User Registration & Login
System:	VendorSetu	**Subsystem:**	Authentication (Module 1 - modules/auth.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026

Short Description: Verify account creation rules (valid / invalid classes, boundaries, several errors at once), login, lock-out after 5 wrong passwords and protection of pages that need a login.

Pre-conditions: Application is running with an empty database. User is on the Register or Log in page of VendorSetu.

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
TC-AUTH-01	Register with valid details Pre: No account exists for 9876543210	System Positive	Name = Ramesh Kumar, Mobile = 9876543210, Role = Street Vendor, Password = ramesh123, Confirm = ramesh123	Registration successful. Please log in. -> Log in page is shown	Registration successful. Please log in. -> Log in page is shown	PASS
TC-AUTH-02	Register with an already registered mobile number Pre: An account already exists for 9876543210	System Negative	Name = Ramesh Kumar, Mobile = 9876543210 (already registered), Password = ramesh123	Mobile number already registered	Mobile number already registered	PASS
TC-AUTH-03	Register with an invalid mobile number	System Negative (EP)	Mobile = 98765 (5 digits), other fields valid	Enter a valid 10-digit mobile number	Enter a valid 10-digit mobile number	PASS
TC-AUTH-04	Register with a weak password	System Negative (EP)	Password = abc, Confirm = abc, other fields valid	Password must be at least 6 characters with letters and digits	Password must be at least 6 characters with letters and digits	PASS
TC-AUTH-05	Register with non-matching passwords	System Negative	Password = ramesh123, Confirm = ramesh124	Passwords do not match	Passwords do not match	PASS
TC-AUTH-06	Log in with correct credentials Pre: Ramesh is registered as a vendor	System Positive	Mobile = 9876543210, Password = ramesh123	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	PASS
TC-AUTH-07	Account lock after 5 wrong passwords Pre: Ramesh is registered as a vendor	System BVA	Mobile = 9876543210; Password = wrong1 (x5), then ramesh123	Attempts 1-4: Invalid mobile number or password | Attempt 5: Account locked after 5 failed attempts. Contact support | Attempt 6 (correct password): Account locked after 5 failed attempts. Contact support	Attempts 1-4: Invalid mobile number or password | Attempt 5: Account locked after 5 failed attempts. Contact support | Attempt 6 (correct password): Account locked after 5 failed attempts. Contact support	PASS
TC-AUTH-08	Register with the name left empty	System Negative (EP)	Name = (empty), other fields valid	Name is required	Name is required	PASS
TC-AUTH-09	Register with digits as the name	System Negative (EP)	Name = 12345, other fields valid	Name can contain only letters, spaces and dots	Name can contain only letters, spaces and dots	PASS (Run 1: FAIL, DEF-04)
TC-AUTH-10	Boundary value - name length 50 / 51 Pre: No account exists for 9876543210	System BVA	Name = 'A' x 50 characters; then Name = 'A' x 51 characters (fresh database each time)	50 characters: Registration successful. Please log in. | 51 characters: Name must be at most 50 characters	50 characters: Registration successful. Please log in. | 51 characters: Name must be at most 50 characters	PASS (Run 1: FAIL, DEF-04)
TC-AUTH-11	Mobile number - invalid equivalence	System	Mobile = 987654321 (9 digits) /	9 digits: Enter a valid 10-digit mobile	9 digits: Enter a valid 10-digit mobile	PASS

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
	classes	Negative (EP)	98765432101 (11 digits) / 5876543210 (starts with 5) / 98765abcde	number | 11 digits: Enter a valid 10-digit mobile number | Starts with 5: Enter a valid 10-digit mobile number | Letters: Enter a valid 10-digit mobile number	number | 11 digits: Enter a valid 10-digit mobile number | Starts with 5: Enter a valid 10-digit mobile number | Letters: Enter a valid 10-digit mobile number	
TC-AUTH-12	Boundary value - password length 5 / 6 Pre: No account exists for 9876543210	System BVA	Password = ab123 (5 characters); then Password = ab1234 (6 characters)	5 characters: Password must be at least 6 characters with letters and digits | 6 characters: Registration successful. Please log in.	5 characters: Password must be at least 6 characters with letters and digits | 6 characters: Registration successful. Please log in.	PASS
TC-AUTH-13	Password with only letters / only digits	System Negative (EP)	Password = abcdefgh (no digit); then Password = 12345678 (no letter)	Only letters: Password must be at least 6 characters with letters and digits | Only digits: Password must be at least 6 characters with letters and digits	Only letters: Password must be at least 6 characters with letters and digits | Only digits: Password must be at least 6 characters with letters and digits	PASS
TC-AUTH-14	No role / forged role	System Security	Role = (not selected); then Role = ADMIN (not in the list)	Not selected: Invalid role selected | ADMIN: Invalid role selected	Not selected: Invalid role selected | ADMIN: Invalid role selected	PASS
TC-AUTH-15	Several invalid fields at once (combination)	System Combination	(a) all fields empty; (b) Mobile = 98765 and Password = abc together	All empty: Name is required | Mobile + password invalid: Enter a valid 10-digit mobile number	All empty: Name is required | Mobile + password invalid: Enter a valid 10-digit mobile number	PASS
TC-AUTH-16	Log in with a wrong password Pre: Ramesh is registered as a vendor	System Negative	Mobile = 9876543210, Password = ramesh124	Invalid mobile number or password	Invalid mobile number or password	PASS
TC-AUTH-17	Log in with an unregistered mobile number Pre: Ramesh is registered as a vendor	System Negative	Mobile = 9000000001 (not registered), Password = ramesh123	Invalid mobile number or password	Invalid mobile number or password	PASS
TC-AUTH-18	SQL injection in the login form Pre: Ramesh is registered as a vendor	System Security	Mobile = ' OR '1'='1, Password = ' OR '1'='1	Invalid mobile number or password	Invalid mobile number or password	PASS
TC-AUTH-19	Failed-attempt counter resets after a successful login Pre: Ramesh is registered as a vendor	System BVA	Mobile = 9876543210; wrong1 (x4), ramesh123, wrong1 (x4), ramesh123	4 wrong + correct: Welcome, Ramesh Kumar! | 4 more wrong: Invalid mobile number or password | Correct again: Welcome, Ramesh Kumar!	4 wrong + correct: Welcome, Ramesh Kumar! | 4 more wrong: Invalid mobile number or password | Correct again: Welcome, Ramesh Kumar!	PASS
TC-AUTH-20	Open the dashboard without logging in Pre: Nobody is logged in	System Security	URL = /dashboard	Please log in first. -> Log in page is shown	Please log in first. -> Log in page is shown	PASS

Post-conditions: A valid registration stores exactly one user with a hashed password; rejected input stores nothing. After 5 consecutive wrong passwords the account stays locked even for the correct password.

E.3 Vendor Profile & Daily Sales Logging

Test Case #:	TC-SALE-01 to TC-SALE-19	**Test Case Name:**	Vendor Profile & Daily Sales Logging
System:	VendorSetu	**Subsystem:**	Vendor Profile & Sales (Module 2 - modules/vendor.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026

Short Description: Verify the business profile rules and that a vendor can log one valid sales entry per day, while invalid amounts, dates and payment modes are rejected without saving anything.

Pre-conditions: Vendor 'Ramesh Kumar' (9876543210 / ramesh123) is registered and logged in. His business profile is saved (unless the case says otherwise).

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
TC-SALE-01	Log a valid daily sale Pre: No sale logged today	System Positive	Date = 04-11-2026 (today), Amount = 1500, Mode = UPI	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	PASS
TC-SALE-02	Log a sale with zero amount	System BVA	Date = 04-11-2026, Amount = 0, Mode = Cash	Sale amount must be greater than 0	Sale amount must be greater than 0	PASS
TC-SALE-03	Log a sale for a future date	System Negative (EP)	Date = 06-11-2026 (2 days ahead), Amount = 1200, Mode = Cash	Sale date cannot be in the future	Sale date cannot be in the future	PASS
TC-SALE-04	Log a second sale for the same date Pre: A sale for 03-11-2026 is already logged	System Negative	Date = 03-11-2026, Amount = 900, Mode = Cash	Sales already logged for this date	Sales already logged for this date	PASS
TC-SALE-05	Log a sale above the daily limit	System Negative (EP)	Date = 04-11-2026, Amount = 150000, Mode = UPI	Sale amount cannot exceed Rs 1,00,000 per day	Sale amount cannot exceed Rs 1,00,000 per day	PASS
TC-SALE-06	Log a sale before saving the business profile Pre: Ramesh has NOT saved his business profile	System Negative	Date = 04-11-2026, Amount = 1500, Mode = UPI	Complete your vendor profile first	Complete your vendor profile first	PASS
TC-SALE-07	30-day summary on the dashboard Pre: No sales logged yet	System Positive	02-11-2026: 1000, 03-11-2026: 2000, 04-11-2026: 3000	30-day sales Rs 6,000; active days 3/30; average Rs 2,000 per active day	30-day sales Rs 6,000; active days 3/30; average Rs 2,000 per active day	PASS
TC-SALE-08	Log a sale with a negative amount	System Negative (EP)	Date = 04-11-2026, Amount = -500, Mode = Cash	Sale amount must be greater than 0	Sale amount must be greater than 0	PASS
TC-SALE-09	Non-numeric / empty amount	System Negative (EP)	Date = 04-11-2026, Amount = abc; then Amount = (empty)	abc: Sale amount must be a number | Empty: Sale amount must be a number	abc: Sale amount must be a number | Empty: Sale amount must be a number	PASS
TC-SALE-10	Special float values as the amount	System Negative (EP)	Date = 04-11-2026, Amount = nan; then Amount = inf	nan: Sale amount must be a number | inf: Sale amount must be a number	nan: Sale amount must be a number | inf: Sale amount must be a number	PASS (Run 1: FAIL, DEF-01)
TC-SALE-11	Boundary value - upper limit Rs 1,00,000 Pre: No sale logged today	System BVA	Date = 04-11-2026, Amount = 100000; then Amount = 100000.01	100000: Sale logged. Your trust score has been updated. | 100000.01: Sale amount cannot exceed Rs 1,00,000 per day	100000: Sale logged. Your trust score has been updated. | 100000.01: Sale amount cannot exceed Rs 1,00,000 per day	PASS
TC-SALE-12	Boundary value - lower limit Pre: No sale logged today	System BVA	Date = 04-11-2026, Amount = 1; then Amount = -0.01	1: Sale logged. Your trust score has been updated. | -0.01: Sale amount must be greater than 0	1: Sale logged. Your trust score has been updated. | -0.01: Sale amount must be greater than 0	PASS

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
TC-SALE-13	Payment mode other than Cash / UPI (forged request)	System Security	Date = 04-11-2026, Amount = 1500, Mode = CARD; then Mode = (empty)	CARD: Payment mode must be CASH or UPI | Empty: Payment mode must be CASH or UPI	CARD: Payment mode must be CASH or UPI | Empty: Payment mode must be CASH or UPI	PASS
TC-SALE-14	Invalid / empty date	System Negative (EP)	Date = 2026-02-30 (no such day); then Date = (empty); Amount = 1500, Mode = UPI	2026-02-30: Enter a valid date (YYYY-MM-DD) | Empty: Enter a valid date (YYYY-MM-DD)	2026-02-30: Enter a valid date (YYYY-MM-DD) | Empty: Enter a valid date (YYYY-MM-DD)	PASS
TC-SALE-15	Boundary value - back-dated entry 90 / 91 days old Pre: No sales logged	System BVA	Date = 06-08-2026 (90 days ago); then Date = 05-08-2026 (91 days ago); Amount = 1500	90 days old: Sale logged. Your trust score has been updated. | 91 days old: Sale date cannot be more than 90 days old	90 days old: Sale logged. Your trust score has been updated. | 91 days old: Sale date cannot be more than 90 days old	PASS (Run 1: FAIL, DEF-02)
TC-SALE-16	Boundary value - today / tomorrow Pre: No sale logged today	System BVA	Date = 04-11-2026 (today); then Date = 05-11-2026 (tomorrow); Amount = 1500	Today: Sale logged. Your trust score has been updated. | Tomorrow: Sale date cannot be in the future	Today: Sale logged. Your trust score has been updated. | Tomorrow: Sale date cannot be in the future	PASS
TC-SALE-17	Several invalid fields at once (combination)	System Combination	(a) Amount = 0, Date = tomorrow, Mode = CARD; (b) Amount = 1500, Date = tomorrow, Mode = CARD	Amount + date + mode invalid: Sale amount must be greater than 0 | Date + mode invalid: Sale date cannot be in the future	Amount + date + mode invalid: Sale amount must be greater than 0 | Date + mode invalid: Sale date cannot be in the future	PASS
TC-SALE-18	Boundary value - years of vending 60 / 61 / -1 Pre: Ramesh is registered but has NOT saved his business profile	System BVA	Business = Fruits & vegetables, Zone = Sector 14 Market, Years = 60; then 61; then -1	60: Profile saved. | 61: Years of vending must be between 0 and 60 | -1: Years of vending must be between 0 and 60	60: Profile saved. | 61: Years of vending must be between 0 and 60 | -1: Years of vending must be between 0 and 60	PASS
TC-SALE-19	Profile with a mandatory field missing Pre: Ramesh is registered but has NOT saved his business profile	System Negative (EP)	Business type = (empty); then Vending zone = (empty); then Years = abc	No business type: Business type and vending zone are required | No zone: Business type and vending zone are required | Years = abc: Years of vending must be a number	No business type: Business type and vending zone are required | No zone: Business type and vending zone are required | Years = abc: Years of vending must be a number	PASS

Post-conditions: A valid entry is saved and counted in the 30-day summary; a rejected entry changes nothing. A rejected profile is not saved.

E.4 Trust Score Calculation

Test Case #:	TC-TS-01 to TC-TS-14	**Test Case Name:**	Trust Score Calculation
System:	VendorSetu	**Subsystem:**	Alt-Credit Scoring Engine (Module 3 - modules/credit_score.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026

Short Description: Verify the 0-100 Trust Score formula, the risk band and the loan-eligibility cut-off of 40.

Pre-conditions: Formula: Sales (active days / 30 x 40) + Repayment (on-time / due x 40, or 20 if no EMI is due yet) + Peer vouching (min(vouches, 3) / 3 x 10) + Tenure (min(years, 5) / 5 x 10), rounded. Bands: 70-100 LOW RISK, 40-69 MEDIUM RISK, 0-39 HIGH RISK.

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
TC-TS-01	Ideal vendor - maximum observable score Pre: No EMI due yet	Unit Positive	Active days = 30/30, EMIs due = 0, Vouches = 3, Years = 5	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	PASS
TC-TS-02	Boundary value - score exactly 40 Pre: No EMI due yet	Unit BVA	Active days = 15/30, EMIs due = 0, Vouches = 0, Years = 0	20 + 20 + 0 + 0 = 40 -> MEDIUM RISK, eligible	20 + 20 + 0 + 0 = 40 -> MEDIUM RISK, eligible	PASS
TC-TS-03	Boundary value - score just below 40 Pre: No EMI due yet	Unit BVA	Active days = 14/30, EMIs due = 0, Vouches = 0, Years = 0	18.7 + 20 + 0 + 0 = 39 -> HIGH RISK, not eligible	18.7 + 20 + 0 + 0 = 39 -> HIGH RISK, not eligible	PASS
TC-TS-04	Vendor with a partial repayment record	Unit Positive	Active days = 24/30, On-time EMIs = 6 of 12 due, Vouches = 1, Years = 2	32 + 20 + 3.3 + 4 = 59 -> MEDIUM RISK, eligible	32 + 20 + 3.3 + 4 = 59 -> MEDIUM RISK, eligible	PASS
TC-TS-05	Invalid input - active days exceed the window	Unit Negative	Active days = 35, Window = 30	Active days cannot exceed the observation window	Active days cannot exceed the observation window	PASS
TC-TS-06	New vendor sees the score on the dashboard Pre: Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales	System Positive	Mobile = 9812345670, Password = sunita123	Dashboard shows Trust Score 20/100, HIGH RISK	Dashboard shows Trust Score 20/100, HIGH RISK	PASS
TC-TS-07	Minimum score - brand-new vendor Pre: No EMI due yet	Unit BVA	Active days = 0/30, EMIs due = 0, Vouches = 0, Years = 0	0 + 20 + 0 + 0 = 20 -> HIGH RISK, not eligible	0 + 20 + 0 + 0 = 20 -> HIGH RISK, not eligible	PASS
TC-TS-08	Maximum score - perfect record	Unit BVA	Active days = 30/30, On-time EMIs = 12 of 12 due, Vouches = 3, Years = 5	40 + 40 + 10 + 10 = 100 -> LOW RISK, eligible	40 + 40 + 10 + 10 = 100 -> LOW RISK, eligible	PASS
TC-TS-09	Vouches and years above their caps Pre: No EMI due yet	Unit BVA	Active days = 30/30, EMIs due = 0, Vouches = 7 (cap 3), Years = 20 (cap 5)	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	PASS
TC-TS-10	Good sales but no EMI paid on time	Unit Positive	Active days = 30/30, On-time EMIs = 0 of 6 due, Vouches = 0, Years = 5	40 + 0 + 0 + 10 = 50 -> MEDIUM RISK, eligible	40 + 0 + 0 + 10 = 50 -> MEDIUM RISK, eligible	PASS
TC-TS-11	Boundary value - score 69 / 70 (MEDIUM / LOW RISK) Pre: No EMI due yet	Unit BVA	Active days = 29/30, then 30/30; EMIs due = 0, Vouches = 0, Years = 5	29 days: 38.7 + 20 + 0 + 10 = 69 -> MEDIUM RISK, eligible | 30 days: 40 + 20 + 0 + 10 = 70 -> LOW RISK, eligible	29 days: 38.7 + 20 + 0 + 10 = 69 -> MEDIUM RISK, eligible | 30 days: 40 + 20 + 0 + 10 = 70 -> LOW RISK, eligible	PASS
TC-TS-12	Invalid input - negative vouches	Unit Negative	Active days = 30/30, EMIs due = 0, Vouches = -1, Years = 5	Score inputs cannot be negative	Score inputs cannot be negative	PASS
TC-TS-13	Invalid input - on-time EMIs exceed	Unit	Active days = 30/30, On-time	On-time EMIs cannot exceed EMIs	On-time EMIs cannot exceed EMIs	PASS

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
	EMIs due	Negative	EMIs = 5, EMIs due = 3, Vouches = 0, Years = 5	due	due	
TC-TS-14	Score goes up after a sale is logged Pre: Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales	System Positive	Date = 04-11-2026, Amount = 800, Mode = Cash	Before: 20/100 | After logging 1 sale: 21/100, HIGH RISK	Before: 20/100 | After logging 1 sale: 21/100, HIGH RISK	PASS

Post-conditions: Score, band and eligibility are returned / displayed; no stored data is modified.

E.5 Loan Application & Bank Review

Test Case #:	TC-LOAN-01 to TC-LOAN-21	**Test Case Name:**	Loan Application & Bank Review
System:	VendorSetu	**Subsystem:**	Loan Management (Module 4 - modules/loan.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026

Short Description: Verify progressive-tier loan application rules and the bank officer's approve / reject / disburse actions.

Pre-conditions: Vendor 'Ramesh Kumar' (9876543210 / ramesh123): profile with 5 years of vending and a sale logged on each of the last 30 days, so his Trust Score is 40 + 20 + 0 + 10 = 70. Bank officer 'Anil Sharma' (9123456780 / bank123) is registered.

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
TC-LOAN-01	Eligible vendor applies for a Tier-1 loan Pre: Ramesh has no loan	System BVA	Amount = 15000 (Trust Score 70, Tier 1 limit 15,000)	Loan application #1 submitted. Status: PENDING.	Loan application #1 submitted. Status: PENDING.	PASS
TC-LOAN-02	Amount above the tier limit Pre: Ramesh has no loan	System Negative (EP)	Amount = 20000	Amount exceeds Tier 1 limit of Rs 15,000	Amount exceeds Tier 1 limit of Rs 15,000	PASS
TC-LOAN-03	Vendor with a low Trust Score applies Pre: Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales (score 20)	System Negative	Amount = 10000	Trust score 20 is below 40. Not eligible	Trust score 20 is below 40. Not eligible	PASS
TC-LOAN-04	Second application while one is pending Pre: Ramesh's application #1 for Rs 15,000 is PENDING	System Negative	Amount = 5000	You already have an active loan	You already have an active loan	PASS
TC-LOAN-05	Officer rejects without remarks Pre: Ramesh's application #1 is PENDING	System Negative	Remarks = (empty), Decision = Reject	Remarks are required when rejecting	Remarks are required when rejecting	PASS
TC-LOAN-06	Vendor tries to open the bank officer page	System Security	URL = /officer	You are not allowed to open that page.	You are not allowed to open that page.	PASS
TC-LOAN-07	Officer approves and disburses Pre: Ramesh's application #1 is PENDING	System Positive	Decision = Approve, then Disburse	Loan #1 APPROVED. | Loan #1 disbursed. 12 EMIs of Rs 1332.73 scheduled.	Loan #1 APPROVED. | Loan #1 disbursed. 12 EMIs of Rs 1332.73 scheduled.	PASS
TC-LOAN-08	Zero / negative loan amount Pre: Ramesh has no loan	System BVA	Amount = 0; then Amount = -5000	0: Loan amount must be greater than 0 | -5000: Loan amount must be greater than 0	0: Loan amount must be greater than 0 | -5000: Loan amount must be greater than 0	PASS
TC-LOAN-09	Non-numeric / empty loan amount Pre: Ramesh has no loan	System Negative (EP)	Amount = abc; then Amount = (empty)	abc: Loan amount must be a number | Empty: Loan amount must be a number	abc: Loan amount must be a number | Empty: Loan amount must be a number	PASS
TC-LOAN-10	Special float values as the loan amount Pre: Ramesh has no loan	System Negative (EP)	Amount = nan; then Amount = inf	nan: Loan amount must be a number | inf: Loan amount must be a number	nan: Loan amount must be a number | inf: Loan amount must be a number	PASS (Run 1: FAIL, DEF-01)
TC-LOAN-11	Boundary value - Tier-1 limit 15,000 / 15,000.01	System BVA	Amount = 15000; then Amount = 15000.01 (fresh database each	15000: Loan application #1 submitted. Status: PENDING. |	15000: Loan application #1 submitted. Status: PENDING. |	PASS

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
	Pre: Ramesh has no loan		time)	15000.01: Amount exceeds Tier 1 limit of Rs 15,000	15000.01: Amount exceeds Tier 1 limit of Rs 15,000	
TC-LOAN-12	Apply before saving the business profile Pre: Ramesh is registered but has NOT saved his business profile	System Negative	Amount = 10000	Complete your vendor profile first	Complete your vendor profile first	PASS
TC-LOAN-13	Two rules broken at once (combination)	System Combination	(a) Sunita, score 20, Amount = 20000 (low score + above limit); (b) Ramesh with application #1 PENDING, Amount = 20000 (active loan + above limit)	Low score + above limit: Trust score 20 is below 40. Not eligible | Active loan + above limit: You already have an active loan	Low score + above limit: Trust score 20 is below 40. Not eligible | Active loan + above limit: You already have an active loan	PASS
TC-LOAN-14	Review an application that is already approved Pre: Ramesh's application #1 is APPROVED	System Negative	Decision = Reject, Remarks = Changed my mind	Only PENDING applications can be reviewed	Only PENDING applications can be reviewed	PASS
TC-LOAN-15	Disburse an application that is still pending Pre: Ramesh's application #1 is PENDING	System Negative	POST /officer/loan/1/disburse	Only APPROVED loans can be disbursed	Only APPROVED loans can be disbursed	PASS
TC-LOAN-16	Forged decision value Pre: Ramesh's application #1 is PENDING	System Security	Decision = DELETE, Remarks = x	Decision must be APPROVE or REJECT	Decision must be APPROVE or REJECT	PASS
TC-LOAN-17	Review a loan that does not exist Pre: Only application #1 exists	System Negative	POST /officer/loan/99/review, Approve	Loan not found	Loan not found	PASS
TC-LOAN-18	Withdraw a pending application, then apply again Pre: Ramesh's application #1 for Rs 15,000 is PENDING	System Positive	Withdraw #1; then Amount = 12000	Application #1 withdrawn. | Loan application #2 submitted. Status: PENDING.	Application #1 withdrawn. | Loan application #2 submitted. Status: PENDING.	PASS
TC-LOAN-19	Reject with remarks; vendor sees them and can re-apply Pre: Ramesh's application #1 is PENDING	System Positive	Remarks = Sales record too short; then Amount = 15000	Loan #1 REJECTED. | Vendor's loan page shows remarks: Sales record too short | Loan application #2 submitted. Status: PENDING.	Loan #1 REJECTED. | Vendor's loan page shows remarks: Sales record too short | Loan application #2 submitted. Status: PENDING.	PASS
TC-LOAN-20	Customer / municipal officer opens a vendor's loan page Pre: Ramesh's application #1 exists	System Security	URL = /loan/1	Customer: You are not allowed to open that page. | Municipal officer: You are not allowed to open that page.	Customer: You are not allowed to open that page. | Municipal officer: You are not allowed to open that page.	PASS (Run 1: FAIL, DEF-05)
TC-LOAN-21	Vendor opens another vendor's loan page Pre: Ramesh's application #1 exists	System Security	URL = /loan/1	You can only view your own loans.	You can only view your own loans.	PASS

Post-conditions: An accepted application is stored as PENDING; approval and disbursal change its status and create the EMI schedule; rejected requests change nothing. Only the owning vendor and bank officers can open a loan's page.

E.6 EMI Schedule & Repayment

Test Case #:	TC-EMI-01 to TC-EMI-18	**Test Case Name:**	EMI Schedule & Repayment
System:	VendorSetu	**Subsystem:**	Loan Management - Repayment (Module 4 - modules/loan.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026

Short Description: Verify the EMI schedule, the repayment rules, loan closure (next tier unlocked) and the default check.

Pre-conditions: Ramesh's Tier-1 loan #1 of Rs 15,000 at 12% p.a. for 12 months has been approved and disbursed today by the bank officer (unless the case says otherwise). Hand calculation: r = 12% / 12 = 0.01, EMI = 15000 x 0.01 x 1.01^12 / (1.01^12 - 1) = Rs 1,332.73.

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
TC-EMI-01	View the EMI schedule	System Positive	Loan #1: P = 15000, r = 12% p.a., n = 12, disbursed 04-11-2026	12 installments; EMI #1 Rs 1,332.73 due 2026-12-04; EMI #12 due 2027-11-04	12 installments; EMI #1 Rs 1,332.73 due 2026-12-04; EMI #12 due 2027-11-04	PASS
TC-EMI-02	Pay EMI #1 before its due date Pre: No EMI paid yet	System Positive	Installment = 1, Amount = 1332.73, paid on 04-11-2026	EMI #1 paid on time.	EMI #1 paid on time.	PASS
TC-EMI-03	Pay less than the EMI amount Pre: No EMI paid yet	System Negative	Installment = 1, Amount = 1000	Amount is less than EMI due of Rs 1332.73	Amount is less than EMI due of Rs 1332.73	PASS
TC-EMI-04	Skip an installment (forged request) Pre: EMI #1 is paid; #2 is due	System Security	POST /vendor/loan/1/pay/3, Amount = 1332.73	Pay earlier installment #2 first	Pay earlier installment #2 first	PASS
TC-EMI-05	Pay the same installment twice (forged request) Pre: EMI #1 is already paid	System Security	POST /vendor/loan/1/pay/1, Amount = 1332.73	Installment already paid	Installment already paid	PASS
TC-EMI-06	Repay all 12 EMIs (progressive lending) Pre: No EMI paid yet	System Positive	12 payments, each equal to the EMI shown	EMI #12 paid on time. Loan fully repaid - you are now eligible for the next tier! -> Dashboard shows eligible limit Rs 25,000 (Tier 2)	EMI #12 paid on time. Loan fully repaid - you are now eligible for the next tier! -> Dashboard shows eligible limit Rs 25,000 (Tier 2)	PASS
TC-EMI-07	Default check marks a 90+ days overdue loan Pre: Loan #1 was disbursed on 07-06-2026 (150 days ago) and no EMI has been paid	System Positive	(button click, no data)	Default check complete: 1 loan(s) marked DEFAULTED.	Default check complete: 1 loan(s) marked DEFAULTED.	PASS
TC-EMI-08	Special float values as the payment Pre: No EMI paid yet	System Negative (EP)	Installment = 1, Amount = nan; then Amount = inf	nan: Payment amount must be a number | inf: Payment amount must be a number	nan: Payment amount must be a number | inf: Payment amount must be a number	PASS (Run 1: FAIL, DEF-03)
TC-EMI-09	Non-numeric / empty payment Pre: No EMI paid yet	System Negative (EP)	Installment = 1, Amount = abc; then Amount = (empty)	abc: Payment amount must be a number | Empty: Payment amount must be a number	abc: Payment amount must be a number | Empty: Payment amount must be a number	PASS
TC-EMI-10	Boundary value - 1 paisa less than the EMI / exact EMI Pre: No EMI paid yet	System BVA	Installment = 1, Amount = 1332.72; then Amount = 1332.73 (fresh database each time)	1332.72: Amount is less than EMI due of Rs 1332.73 | 1332.73: EMI #1 paid on time.	1332.72: Amount is less than EMI due of Rs 1332.73 | 1332.73: EMI #1 paid on time.	PASS
TC-EMI-11	Zero / negative payment Pre: No EMI paid yet	System Negative (EP)	Installment = 1, Amount = 0; then Amount = -1332.73	0: Amount is less than EMI due of Rs 1332.73 | -1332.73: Amount is less than EMI due of Rs 1332.73	0: Amount is less than EMI due of Rs 1332.73 | -1332.73: Amount is less than EMI due of Rs 1332.73	PASS
TC-EMI-12	Pay an EMI after its due date Pre: Loan #1 was disbursed on 25-09-	System Positive	Installment = 1, Amount = 1332.73, paid on 04-11-2026	EMI #1 paid late. -> Trust Score 50/100	EMI #1 paid late. -> Trust Score 50/100	PASS

TC ID	Test Scenario	Type	Input (Test Data)	Expected Output	Actual Output	Remark
	2026 (40 days ago), so EMI #1 is overdue					
TC-EMI-13	Paying on time raises the Trust Score Pre: No EMI paid yet; Trust Score 70	System Positive	Installment = 1, Amount = 1332.73	Before: 70/100 | After paying EMI #1 on time: 90/100	Before: 70/100 | After paying EMI #1 on time: 90/100	PASS
TC-EMI-14	Installment number that does not exist (forged request) Pre: No EMI paid yet	System Security	POST /vendor/loan/1/pay/13 and /pay/0, Amount = 1332.73	#13: Installment not found | #0: Installment not found	#13: Installment not found | #0: Installment not found	PASS
TC-EMI-15	Vendor pays another vendor's EMI (forged request) Pre: Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales	System Security	POST /vendor/loan/1/pay/1, Amount = 1332.73	You can only pay your own loans	You can only pay your own loans	PASS (Run 1: FAIL, DEF-06)
TC-EMI-16	Pay an EMI of a loan that does not exist Pre: Only loan #1 exists	System Negative	POST /vendor/loan/99/pay/1, Amount = 1332.73	Loan not found	Loan not found	PASS (Run 1: FAIL, DEF-06)
TC-EMI-17	Pay an EMI of an approved but not disbursed loan Pre: Ramesh's loan #1 is APPROVED (not disbursed)	System Negative	POST /vendor/loan/1/pay/1, Amount = 1332.73	Loan is not active for repayment	Loan is not active for repayment	PASS
TC-EMI-18	Boundary value - EMI overdue 90 / 91 days Pre: Loan #1 is DISBURSED, no EMI paid	System BVA	EMI #1 due date set to 90 days ago; then to 91 days ago (fresh database each time)	90 days: Default check complete: 0 loan(s) marked DEFAULTED. | 91 days: Default check complete: 1 loan(s) marked DEFAULTED.	90 days: Default check complete: 0 loan(s) marked DEFAULTED. | 91 days: Default check complete: 1 loan(s) marked DEFAULTED.	PASS

Post-conditions: Paid installments are marked PAID (on time / late); when all 12 are paid the loan is CLOSED and Tier 2 (Rs 25,000) is unlocked.

E.7 Test Summary

Module	Test cases	Passed	Failed	Pass rate
User Registration & Login	20	20	0	100%
Vendor Profile & Daily Sales Logging	19	19	0	100%
Trust Score Calculation	14	14	0	100%
Loan Application & Bank Review	21	21	0	100%
EMI Schedule & Repayment	18	18	0	100%
Total	**92**	**92**	**0**	**100%**

Integration test (tests/test_integration.py)	What it verifies	Result
test_end_to_end_loan_flow	Register vendor & officer → login → profile → sale → apply ₹15,000 → approve → disburse (12 EMIs of ₹1,332.73) → pay EMI #1 on time	PASS
test_role_based_access	Customer cannot open the officer page; logged-out user is sent to login	PASS

Actual outputs above are recorded by executing the code, not typed by hand; re-running tests/run_test_cases.py and docs/build_srs.py regenerates these tables.

E.8 Defects Found by Testing

Run 1 executed the same cases on the first version of the code (git commit 0ce1911): 9 cases FAILED. Each was logged as a defect, fixed and re-tested (Run 2, results above). The full Run-1 outputs are in Section 9 of the Test Case Document.

Defect	Severity	Found by	What happened in Run 1	Fix	Re-test
DEF-01	High	TC-SALE-10, TC-LOAN-10	Typing 'nan' as a sale or loan amount crashed the app (HTTP 500 Internal Server Error); 'inf' as a loan amount gave a misleading message.	New helper parse_number() in database.py rejects values that are not finite numbers.	PASS - closed
DEF-02	Medium	TC-SALE-15	A sale dated 1900-01-01 (or any old date) was accepted.	vendor.log_daily_sale() rejects entries more than 90 days old (MAX_BACKDATE_DAYS); FR-VEN-5 updated.	PASS - closed
DEF-03	High	TC-EMI-08	Paying an EMI with amount 'nan' or 'inf' marked the installment PAID.	loan.pay_emi() uses parse_number().	PASS - closed
DEF-04	Low	TC-AUTH-09, TC-AUTH-10	Names such as '12345' or a 300-character string were accepted at registration.	auth.validate_name(): letters, spaces, dots and apostrophes only, at most 50 characters; FR-AUTH-1 updated.	PASS - closed
DEF-05	High	TC-LOAN-20	A customer or municipal officer could open any vendor's loan page and see the amount and EMIs.	Only the owning vendor and bank officers may open a loan page (LOAN_VIEW_ROLES in app.py).	PASS - closed
DEF-06	Low	TC-EMI-15, TC-EMI-16	Paying an EMI of a loan that does not exist, or of another vendor's loan, showed two error messages ('Loan not found | Loan not found', 'You can only pay your own loans | You can only view your own loans.').	The pay_emi route redirects to the dashboard when the loan is missing or not the vendor's own.	PASS - closed